Conditional generation of visible language

a Training

60 visible glyph images
Frozen causal reader

→

Continuous visual history
384 state + 1024 image field

History conditions every residual scale

Four noisy future cells
Independent or shared noise levels

→

Conditional strip denoiser
FiLM and cell attention

→

Velocity and visual losses
5.90M trainable parameters

b Inference

Noise and visible history

→

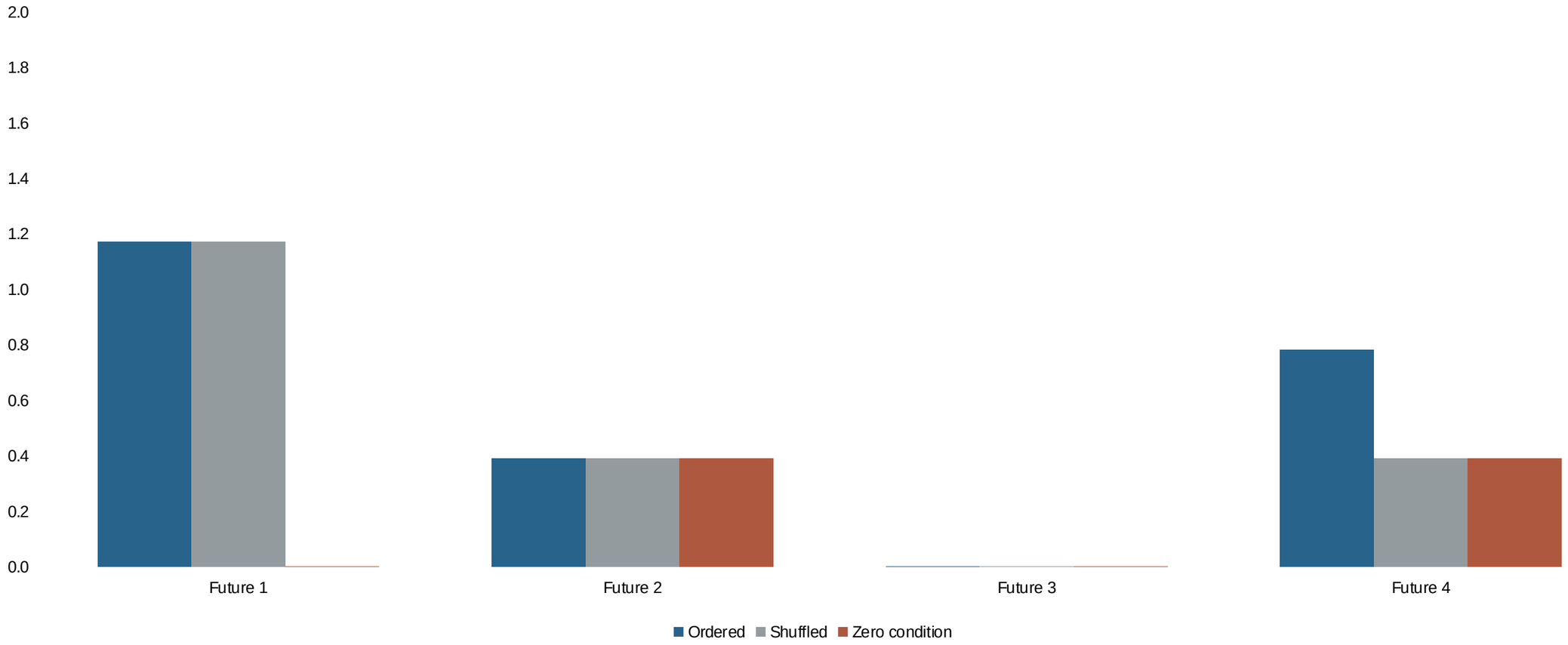
32-level reverse process

→

Four output images
Append pixels to next context

Source: V53 frozen protocol, implementation and locked audit. Schematic objects are editable.

Direct block generation under paired visual histories



256 prefixes, one primary sample each, four output cells
Ordered block identity 0.586%, shuffled 0.488%, zero condition 0.195%

Source: locked V53 development report. Native chart includes its editable evidence workbook.

Unselected output images expose the language error



Fixed first eight audit rows. Pixels are model output, with nearest-neighbour enlargement. No glyph replacement.

Exact raster likelihood preserves a visible copy path

64 writing images
Frozen 16.28M reader

→

Continuous visual state
Hidden384 + image field1024

→

Fixed inverse field
Aligned signed pixel plan

Training: exact Bernoulli image NLL

Earlier target rows
Four-layer row transformer

→

Earlier pixels within row
Strictly masked MADE

→

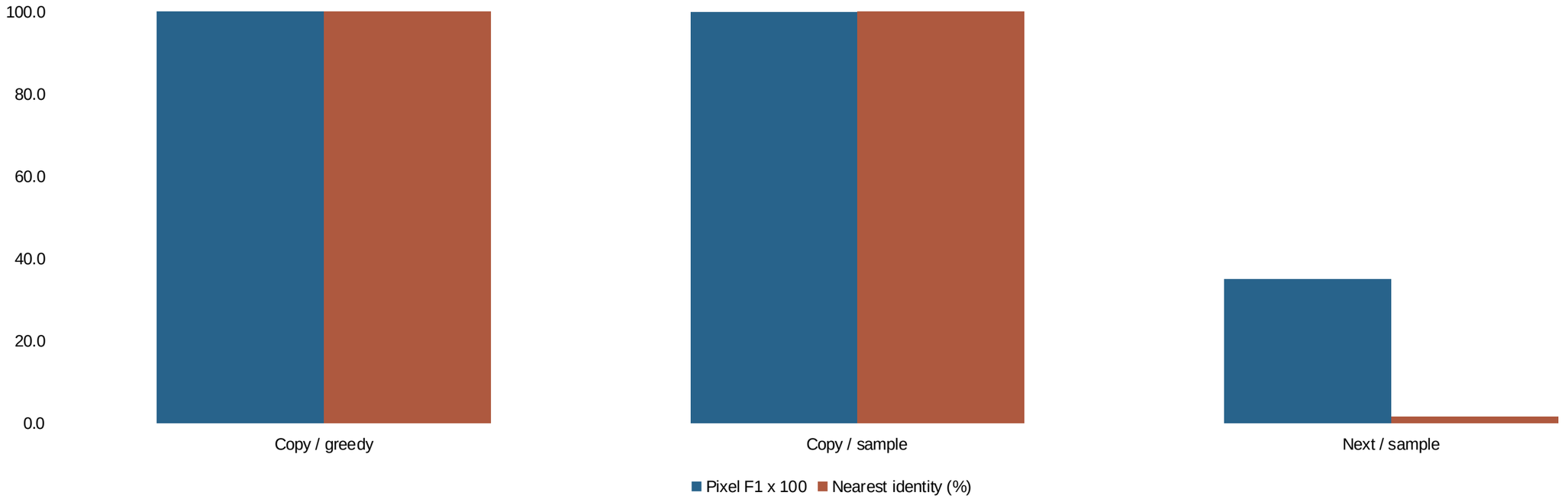
Aligned logit + residual
Normalized next-image law

25% supplied-image motor exercises; 75% natural predicted conditions. No glyph inventory.

Inference: sample 1,024 causal pixels; append the visible raster to history

2.43M new density parameters; 18.71M total. Supplied-image copying is architectural, not language evidence.

Copying succeeds; autonomous language remains unresolved

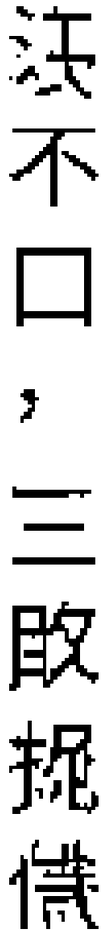


Exact-image copying: 256/256 greedy. Autonomous target identity: 4/256 ordered, 3/256 shuffled.

Target NLL: 15.219 ordered; 15.254 shuffled nats/image
Paired advantage 0.0355 nat; source-record bootstrap 95% CI [-0.1014, 0.1693]

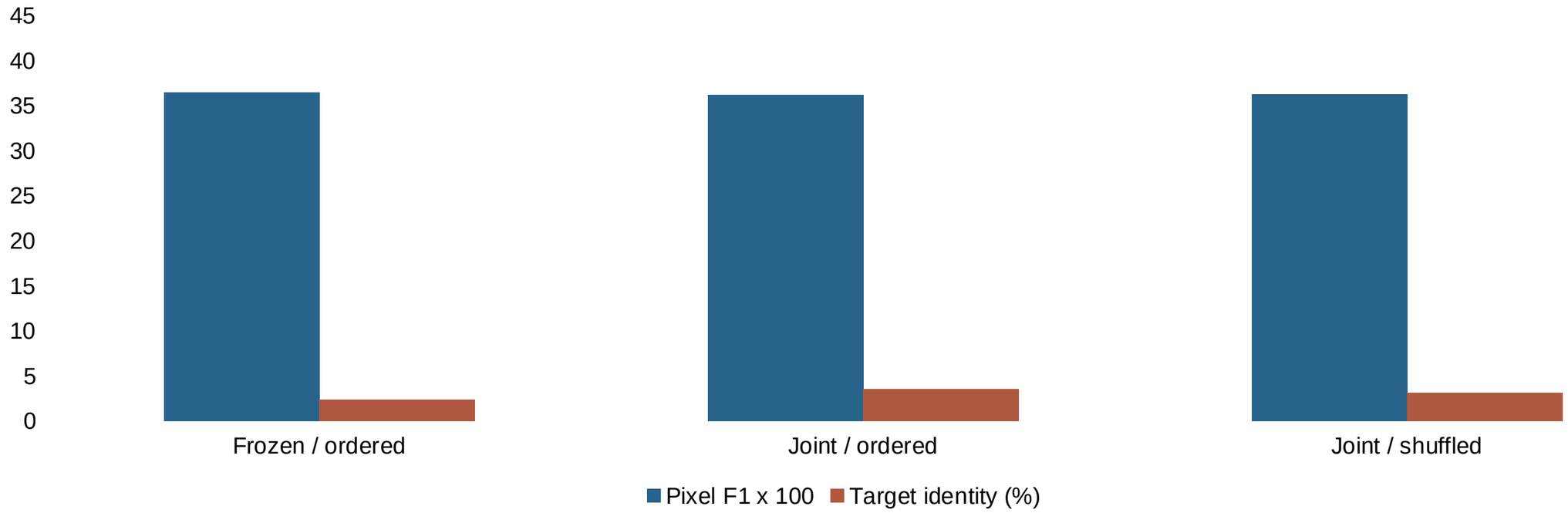
V54 locked development audit. Native chart and embedded workbook. Supplied-image controls reveal the answer.

Actual rasters: exact motor copying and failed continuation

Target	Supplied / greedy	Ordered / sample	Shuffled / sample	Ordered / greedy
				

First eight fixed audit rows; nearest-neighbour enlargement. No OCR, image replacement or selected examples.

Matched joint learning: better likelihood, incomplete generation



Exact target NLL: joint 7.026; matched frozen 7.579 nats/image
Paired gain 0.553; 95% source-record interval [0.450, 0.665]

Both arms retain exact supplied-image copying. Joint model passes 6/8 gates; generation remains unqualified.

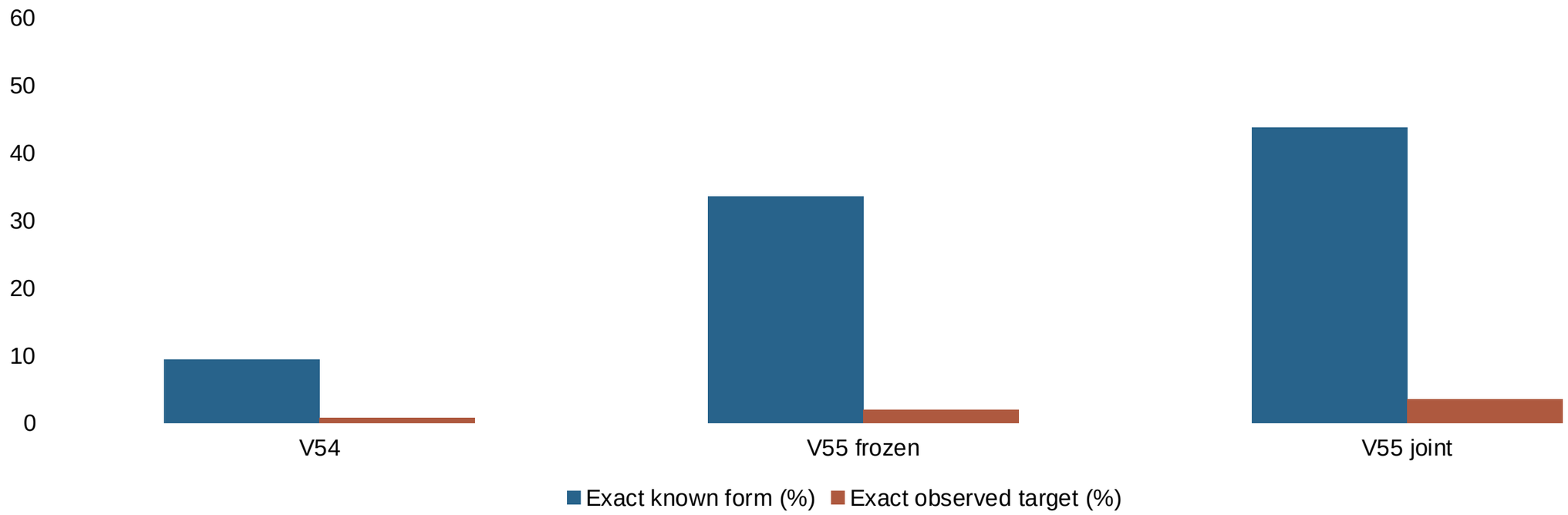
V55 locked matched audit; 512 NLL targets, 256 single-sample prefixes. Native chart includes its editable workbook.

Raw matched output: exact copying and remaining language errors

Target	Frozen sample	Joint sample	Joint shuffled	Joint greedy	Supplied copy
江	疫	江	江	江	江
歲	亟	亟	亟	，	歲
般	鴿	一	一	一	般
精	追	些	些	，	精
六	四	四	四	三	六
他	路	路	路	不	他
王	用	苯	抂	王	王
物	蒴	恭	恭	人	物

First eight fixed audit rows, no selection. Pixels are actual output; no OCR, evaluator-bank replacement or external inference.

Clear glyphs and correct language are separate tests

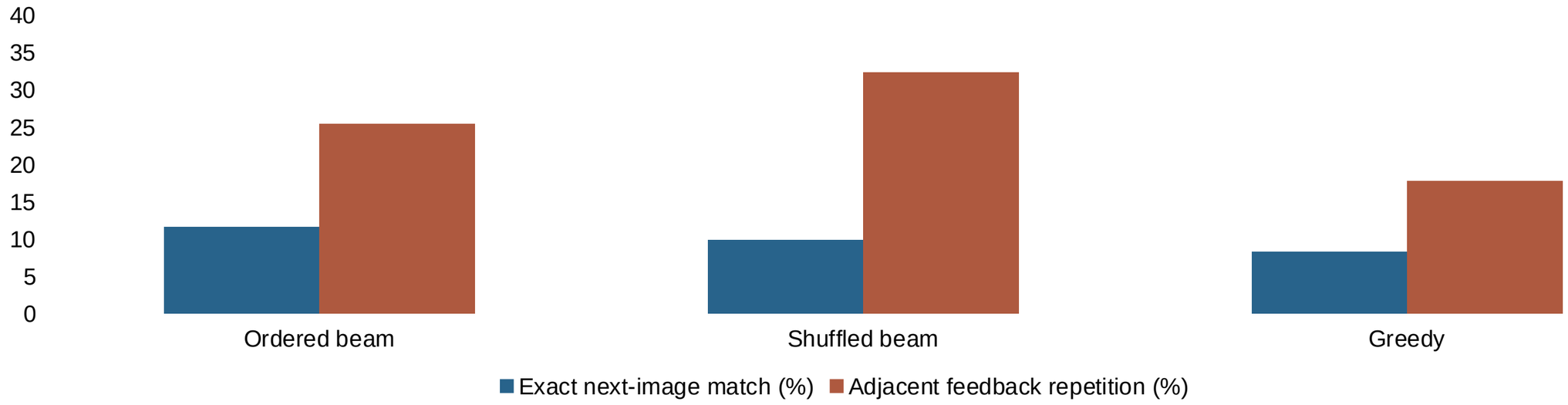


Exact known-form gain, joint minus frozen: 10.156 percentage points
95% source-record interval [4.219, 16.240]; 256 saved first samples

The fixed 9,932-image bank is used only for evaluation. No image is repaired or replaced.

Post-result diagnostic; all original qualification decisions remain unchanged. Native chart includes its editable workbook.

Raw-pixel beam search: better immediate choice, repetitive output

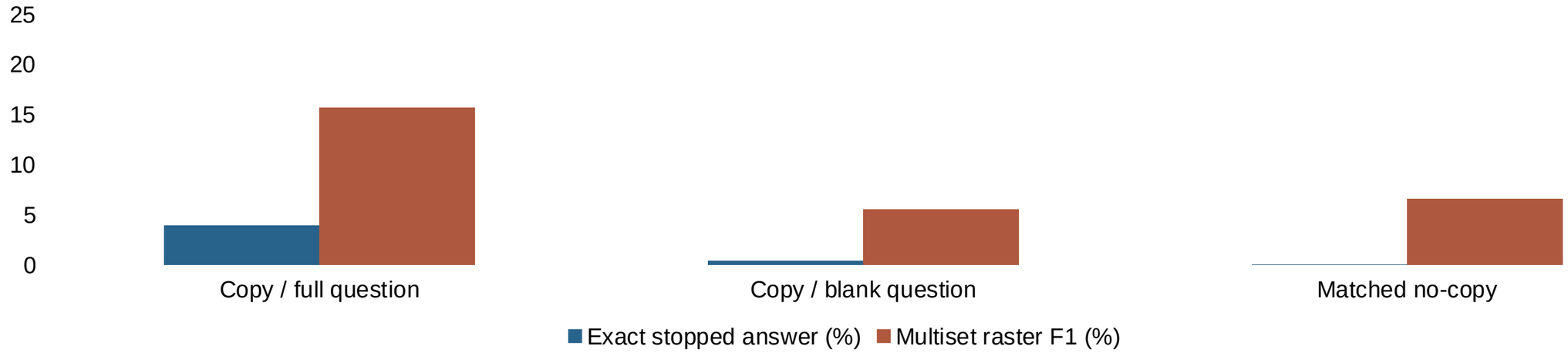


2,048 prefixes: beam gain over greedy 3.369 points [2.351, 4.412]; over shuffled 1.758 [0.966, 2.552]
Observed-stream repetition: 0.112%; beam: 25.446%. Rollout gate fails; decoder rejected.

Observed future	Ordered beam	Shuffled beam	Original greedy
不尬的就跑回来了	一个，不是一个，	一个，不不不不不	，不不一，不不不
的女婿。丈人教我	，不是一人，不是	，不不不不不不不	，不不不不不不不

Unselected first two eight-step audit streams. Unchanged V55 weights; no vocabulary or repair. Both controls use highest FP32 precision.

Natural Chinese questions: some binding, inaccurate complete answers



Question-dependent F1 gain: 10.165 points [7.043, 13.371]
Exact answers: 10/256; both questions correct: 1/393 pairs. Four of seven gates pass; model rejected.

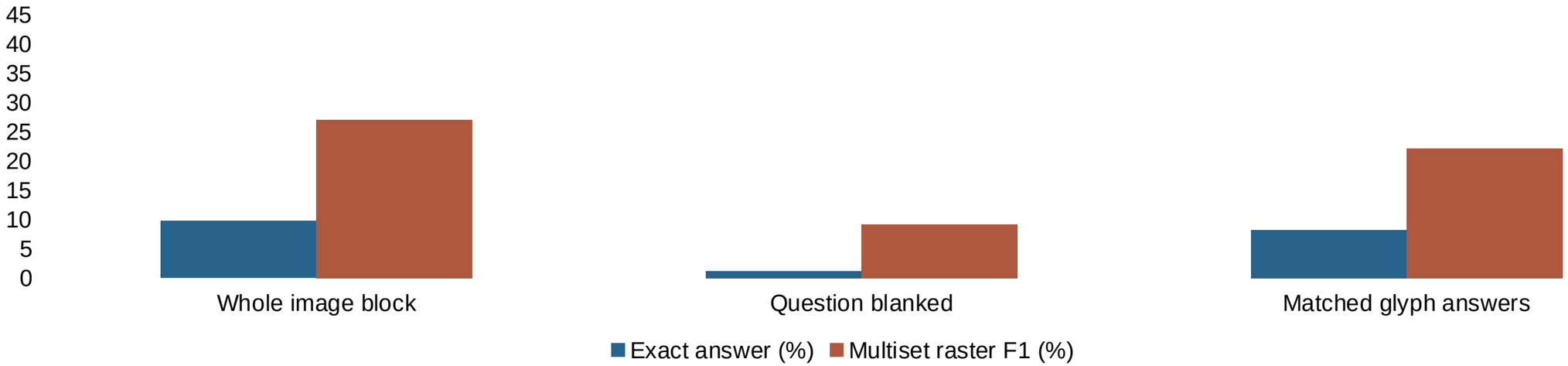
Q1 １９４０年８月世宗大学更名成什么？

Q2 邵伯温是在什么时候去世？

Reference 1	Copy	Blank question	No-copy
首尔女子师范学院 高宗绍兴四年（１１３４年）	１９４０年５月，Ｙ月世宗大学” 邵溥、邵博、邵传后》	节居绍圣、利州路转运作）	１９３０年 １３３０年

First two fixed audit rows; actual output images. White means zero emitted cells. All primary copy-arm answer cells use observed prompt images.

Whole observed image answers: better F1, unresolved exact-match gain



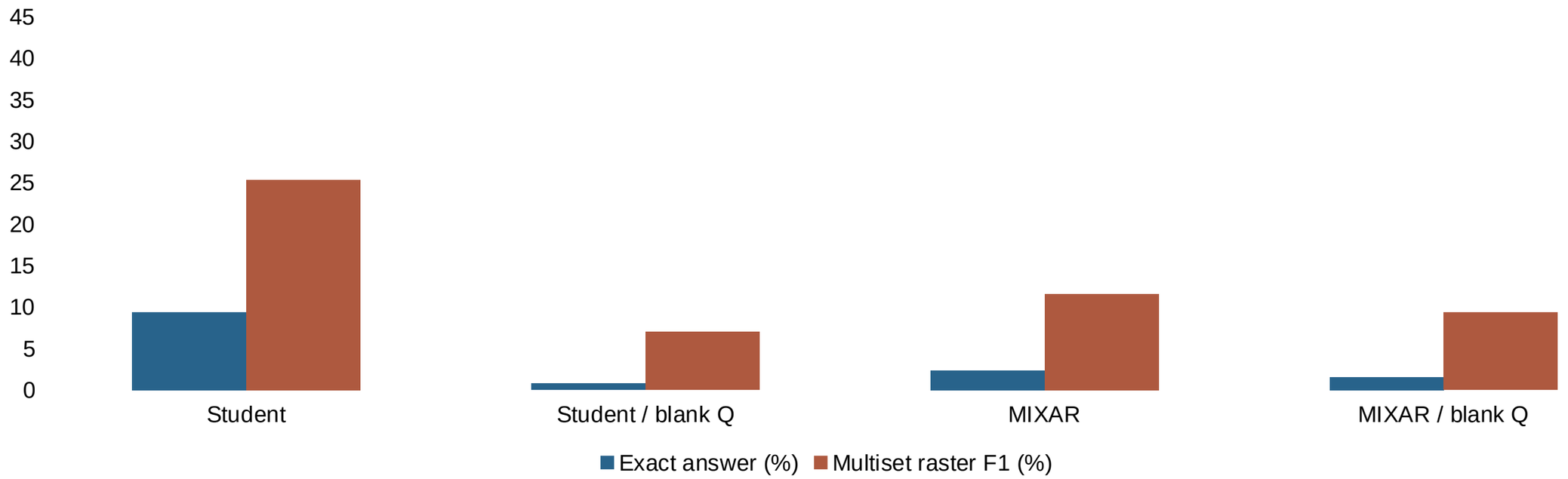
25/256 exact answers; matched glyph control 21/256. F1 gain 4.893 points [0.469, 9.047].
Only 4/393 paragraph pairs have both answers correct. Six of seven gates pass; reading rejected.

Prompt pixels → Causal reader → Start / end law → Source image answer

	Retained reference	Actual block answer
Audit row 1	首尔女子师范学院	“世宗学院”
Audit row 2	高宗绍兴四年（1134年）	卒于高宗绍兴四年（1134年）

First two fixed audit rows. Copying preserves selected pixels; it does not certify the meaning of the selected text.

Frozen visual foundation probe: larger pretrained features do not help here

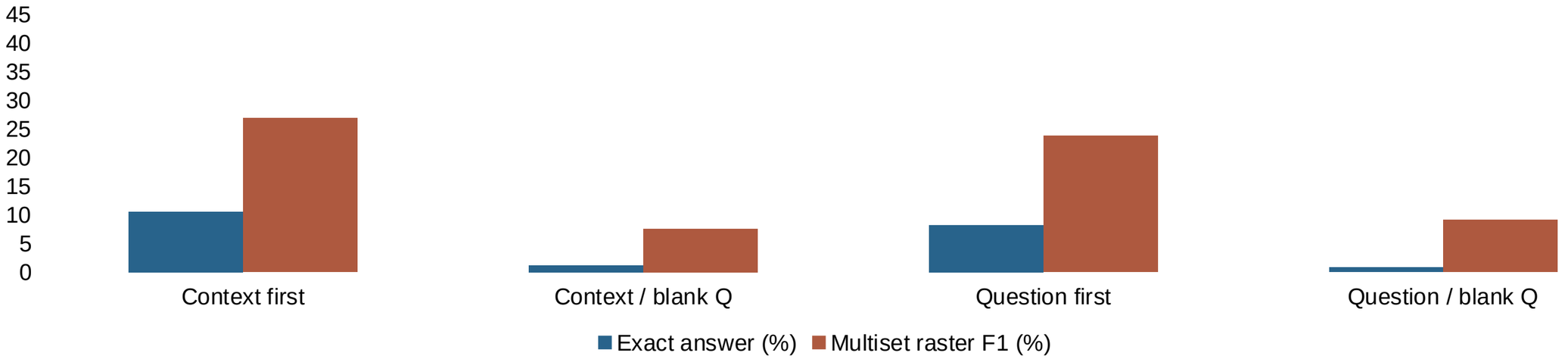


Identical 789,520-parameter heads; 4,000 matched updates each.
Student 24/256 exact; MIXAR 6/256. MIXAR-minus-student F1 −13.752 points [−18.247, −9.094].

MIXAR has different pretraining and native rendering; this probe uses our Noto CJK grid.
External four-H200 pretraining is not one-4090 work. Frozen readout comparison rejected; not a general encoder ranking.

Continuous prompt-image features only. Shared cached release weights; no external inference call or answer annotation enters feature preparation.

Does reading the question first improve evidence selection?



Context first

Passage images X → Question images Q → Answer

Question first

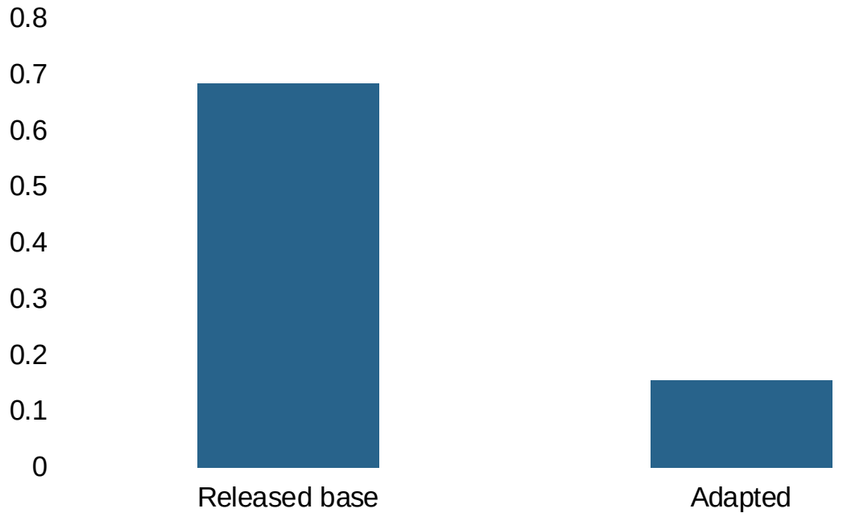
Question images Q → Passage images X → Answer

Question-first minus context-first F1: -3.143 points [-6.544, 0.446].
4/8 registered gates pass; decision: reject_question_first_intervention.

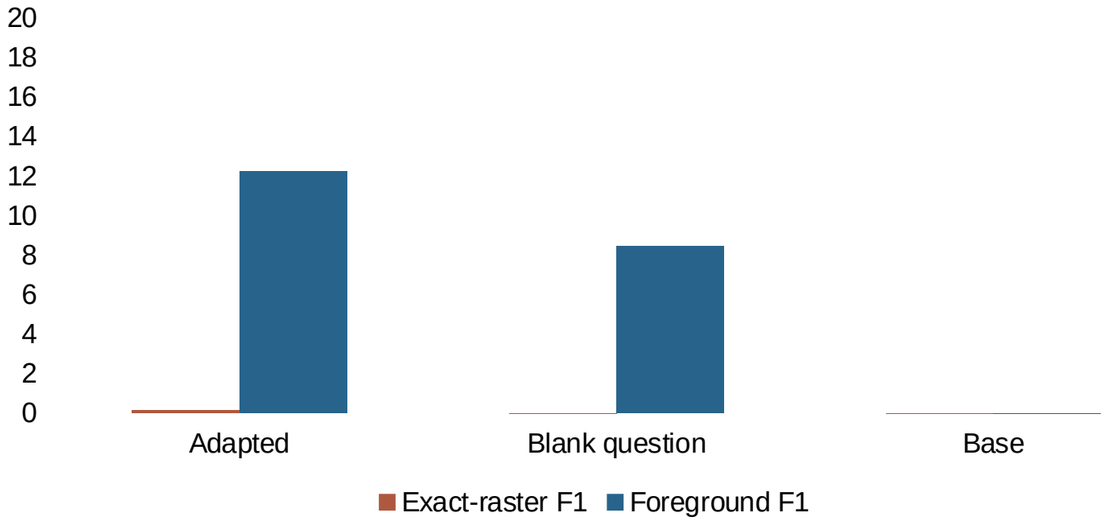
Same checkpoint, data, parameters and training budget. Source pixels only; reused development, not a sealed final test or general language qualification.

Native free writing: improved pixel fit, failed language generation

Held-out answer-pixel NLL (nats)



Generated image scores (%)



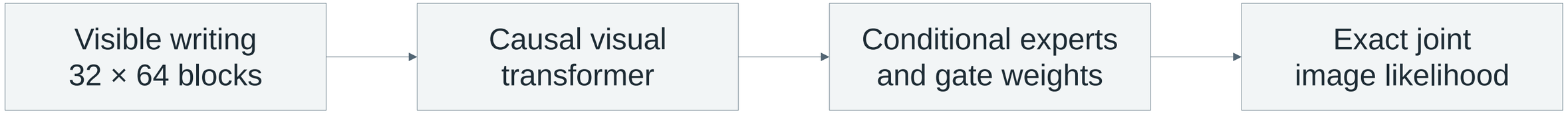
0/256 exact answers and 0/393 pairs both correct. Five of ten registered gates pass.
477.15M externally pretrained parameters, fully adapted for 1,000 updates on GPU 0.

	Retained reference	Actual free-pixel answer
Audit row 1	首尔女子师范学院	
Audit row 2	高宗绍兴四年（１１３４年）	

Training: 1,501.1 s and 9.912 GiB. Offline native-PNG replay passes. External pretraining and reused development disclosed.

Conditional whole-image density

Training: prompt and observed answer images



$$p(y \mid h) = \Sigma k\ \pi k(h)\ \Pi i\ \text{Bernoulli}(yi\ ;\ \text{sigmoid}(aki(h)))$$

One shared expert choice couples all pixels in a bitmap. Each expert is a neural function of context.
K = 1: 477.15M parameters. K = 8: 490.00M. Same examples and update budget; different head size.

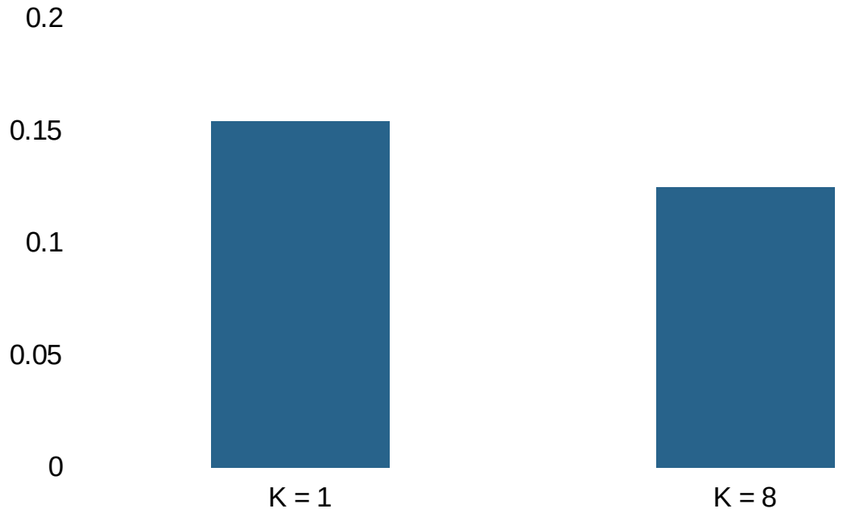
Inference: image prompt → candidate bitmaps → score under full density → physical pixels

Feed generated pixels back into the visual context; stop at an all-white image block.
Component-mode search is approximate MAP. No character bank, OCR, Unicode IDs or external LM in the actor.

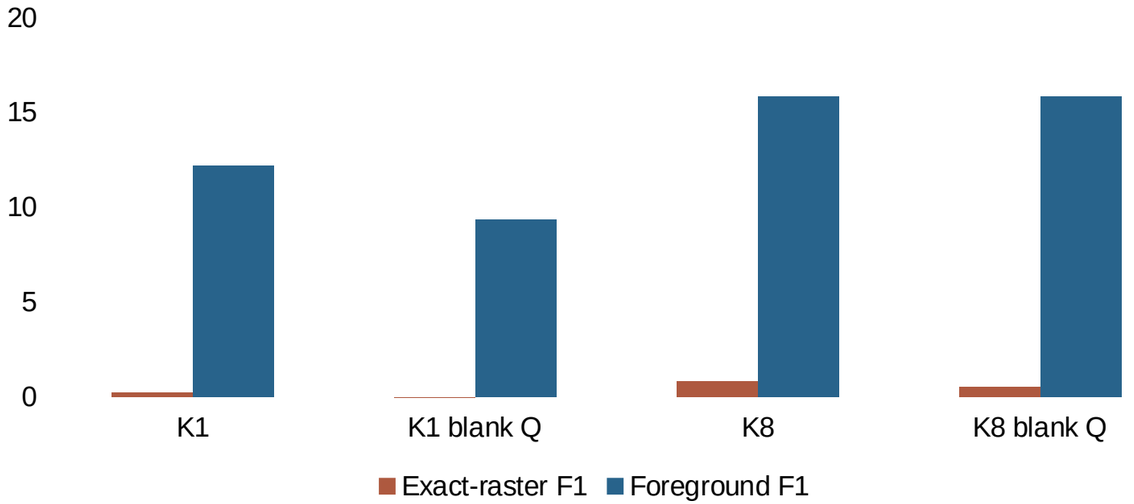
Externally pretrained MIXAR foundation. The mixture law is a tested hypothesis, not a guarantee of readable language.

Conditional density: held-out fit and generated writing

Held-out answer-pixel NLL (nats)



Generated image scores (%)



Exact answers K1 / K8: 0 / 0 of 256. Both-correct pairs: 0 / 0 of 393.
Registered decision: reject density intervention. General image-language quality remains unqualified.

	Retained reference	K = 1	K = 8
Audit row 1	首尔女子师范学院	[illegible]	１９４０年
Audit row 2	高宗绍兴四年（１１３４年）	[illegible]	１９[illegible]３[illegible]０[illegible]１

First two fixed primary answers, unrepaired; white display padding. One seed, reused development, external pretraining disclosed.

Question-conditioned image-language training

a Observed training images: two questions about one paragraph

Question region Q0

哪个人是世纪越南反唐起义军的军事首领？

Recorded answer Y0 冯兴

Question region Q1

冯兴与他的弟弟冯骇接受谁的建议，率众起义？

Recorded answer Y1 杜英翰

Each full prompt Xi also contains the same paragraph image. Appended white termination blocks are not drawn.

b Four causal likelihoods under the same parameters

(X0, Y0) (X0, Y1)
(X1, Y0) (X1, Y1)

→

Shared image generator
490M parameters, 8 experts

→

	Y0	Y1
X0	e_{00}	e_{01}
X1	e_{10}	e_{11}

eij = mean negative log likelihood of the complete answer suffix, including termination

c Positive image likelihood with paired question assignment

$$D = e_{01} + e_{10} - e_{00} - e_{11}$$

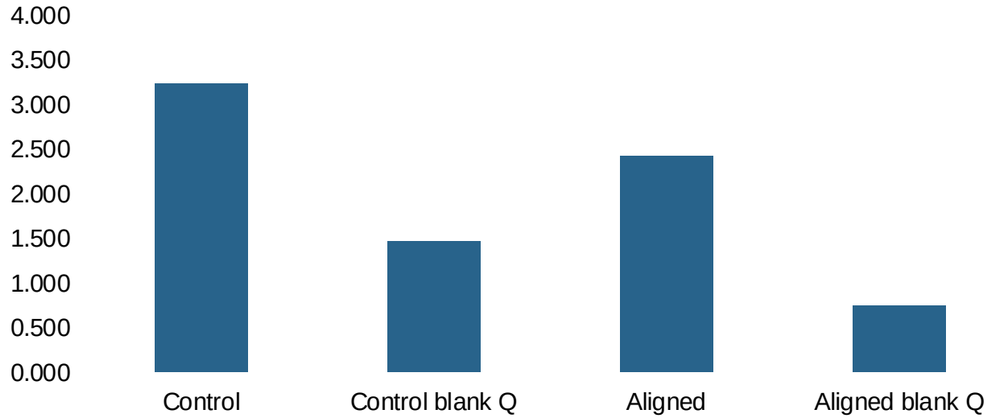
λ = 0: control; λ = 1: assignment

$$L = (e_{00} + e_{11}) / 2 + \lambda \cdot 0.01 \cdot \mathrm{softplus}(-D / 0.01)$$

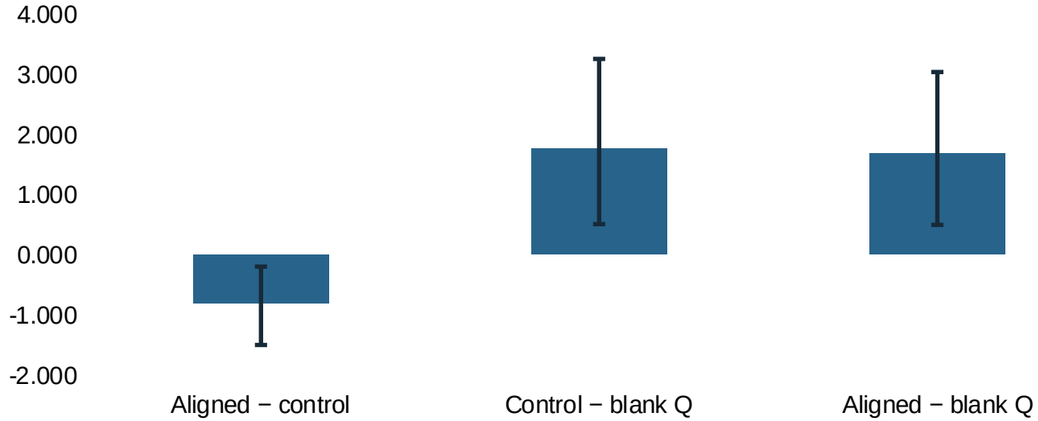
Observed targets, not model outputs. Both arms start from the same V62 EMA; autonomous answers are evaluated separately.

Question assignment: completed free-writing test

Strict-raster F1 (%)



Paired differences with 95% intervals (pp)



Exact answers: control 2/256, aligned 1/256. Both-correct pairs: 0/393 in each arm.

First two fixed primary answers

1: Reference 首尔女子师范学院

1: Control



1: Aligned



2: Reference 高宗绍兴四年（1134年）

2: Control



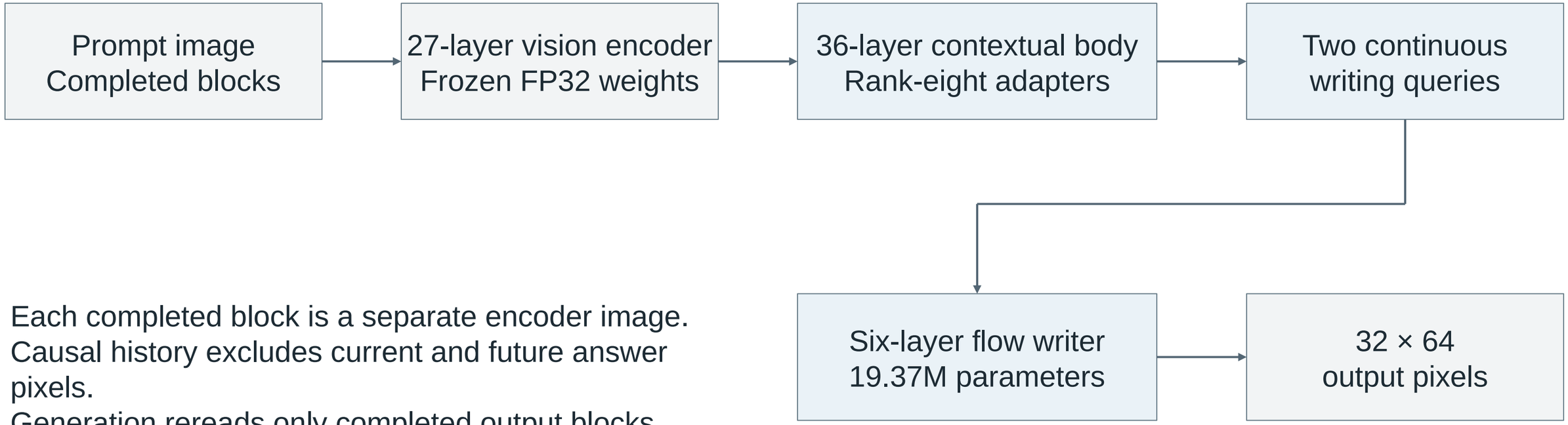
2: Aligned



Unrepaired pixels and white padding. One seed, reused development and external pretraining. Both reading gates fail.

Complete continuous image actor

7.548B total parameters 25.58M trainable 17.181 GiB peak allocated



Each completed block is a separate encoder image.
Causal history excludes current and future answer pixels.
Generation rereads only completed output blocks.

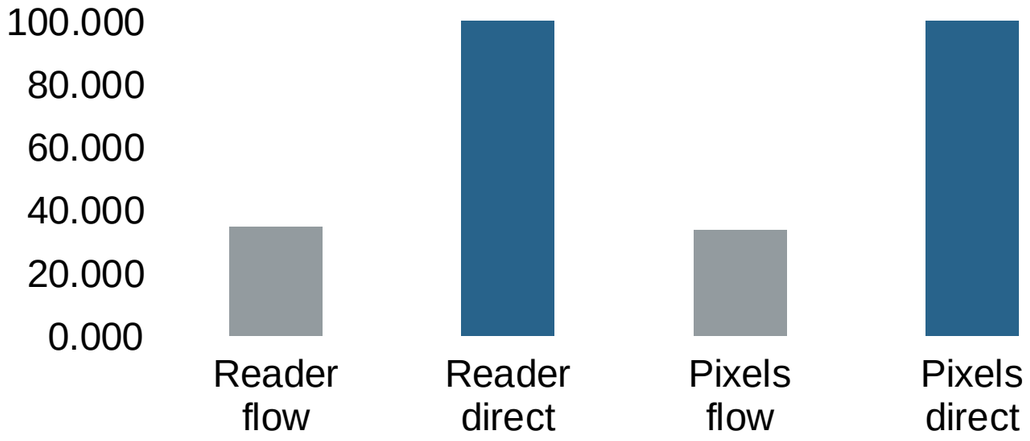
$$Yt = (1 - t)\,U + t\,Y \qquad U \sim N(0, I) \qquad Lflow = E\,||v(Yt,t,h) - (Y - U)||^2$$

Frozen contextual weights use BF16 storage with FP32 computation. A separate classifier predicts stopping.

Execution checks pass. Writing and language quality remain unqualified.

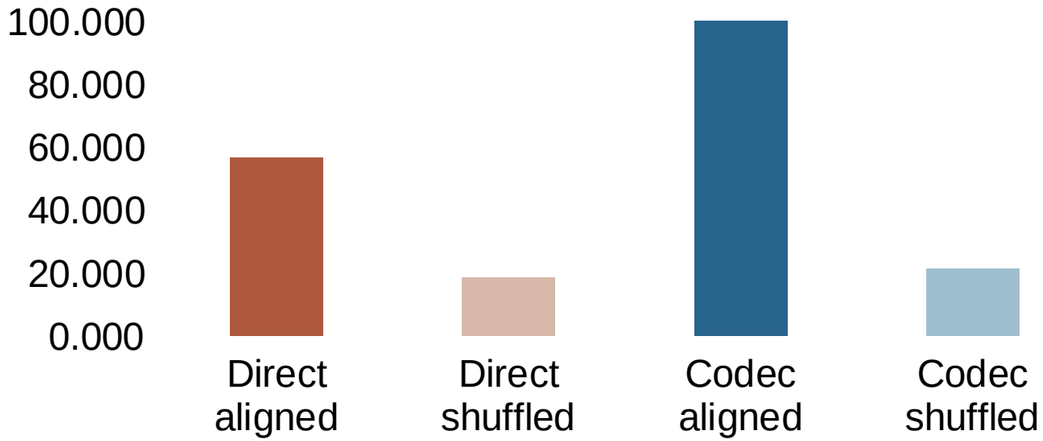
Writing fidelity depends on the output representation

a Training F1 (%) | n=64



Exact blocks: 0/64, 63/64, 0/64, 64/64

b Development F1 (%) | n=128



Exact aligned: direct 3/128; fixed codec 126/128

c First two fitting rows

	Reference	Reader flow	Reader direct	Pixels flow	Pixels direct
1					
2					

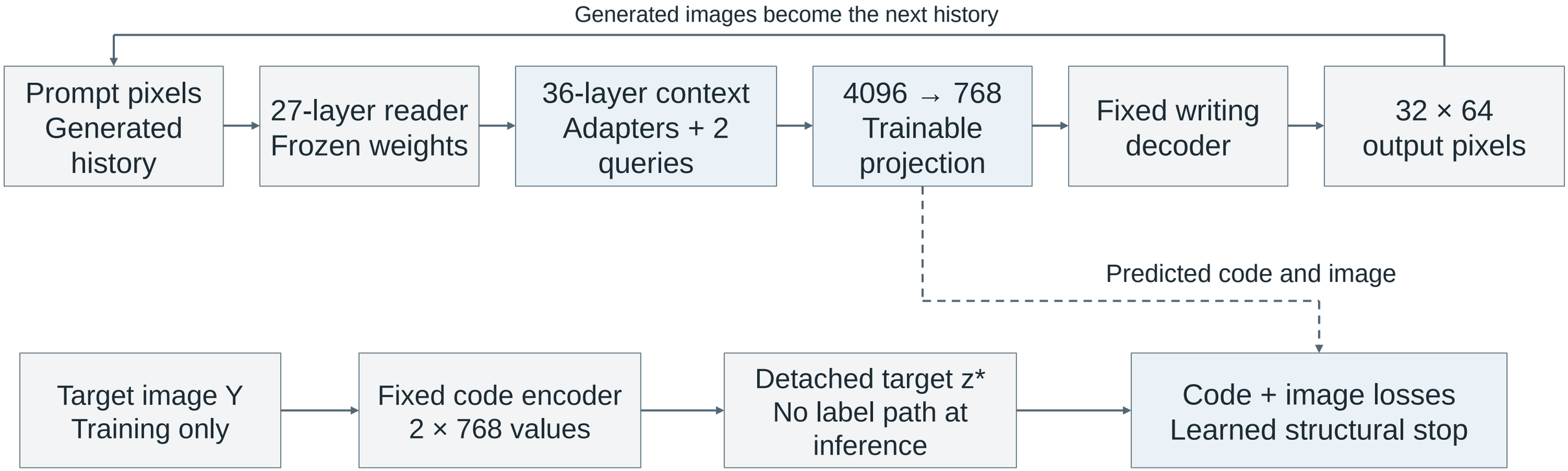
d First two development rows

	Reference	Direct	Codec
1			
2			

Fitting uses training images. Development compares supplied representations with unequal prior training; neither is question answering.

Reading and writing use separate continuous spaces

7.539B total parameters 9.352M trainable 16.953 GiB peak allocated



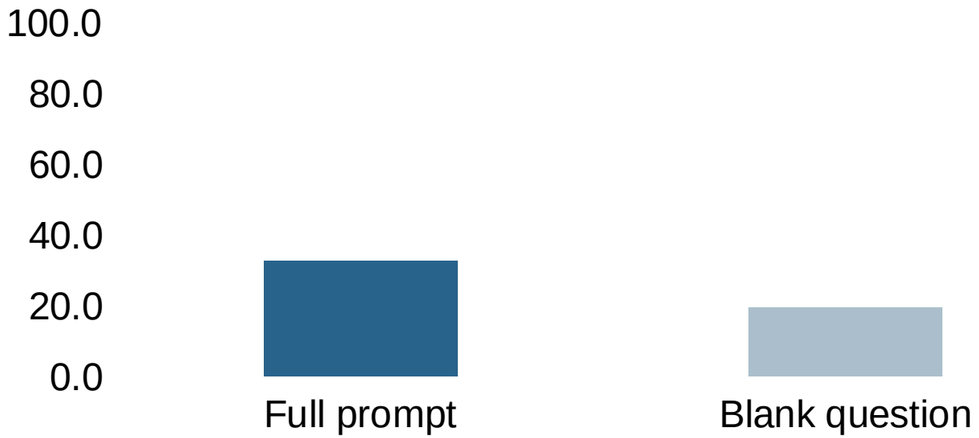
$$L = MSE(N(\hat{z}), N(z^*)) + MSE(D(\hat{z}), Y) + Lstop$$

Frozen decoder weights still pass gradients into the predicted codes and contextual adapters.

All 24 complete-model execution checks pass. Two-update output is fragmented; autonomous language remains unqualified.

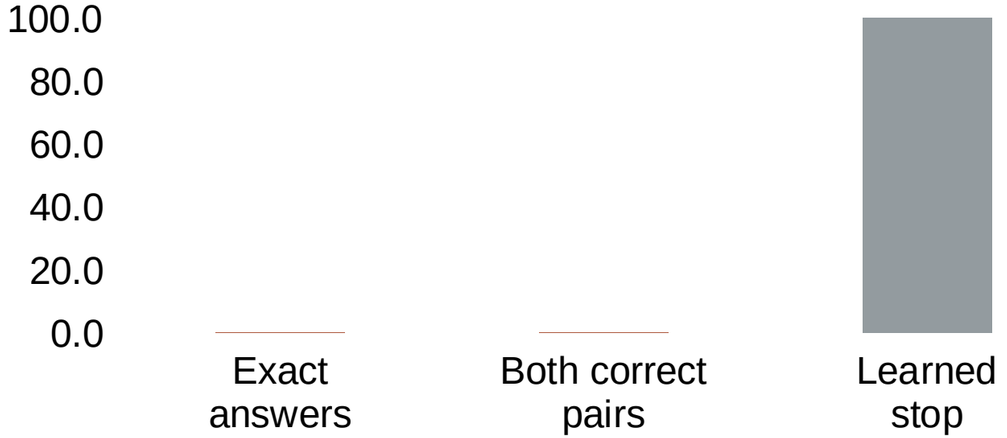
Question effects persist when accurate writing fails

a Chinese foreground F1 (%) | n=128



Paired gain: 13.16 points [10.93, 15.73]

b Exact answers and learned stopping (%)



0/128 exact answers; 0/64 complete pairs; 128/128 stops

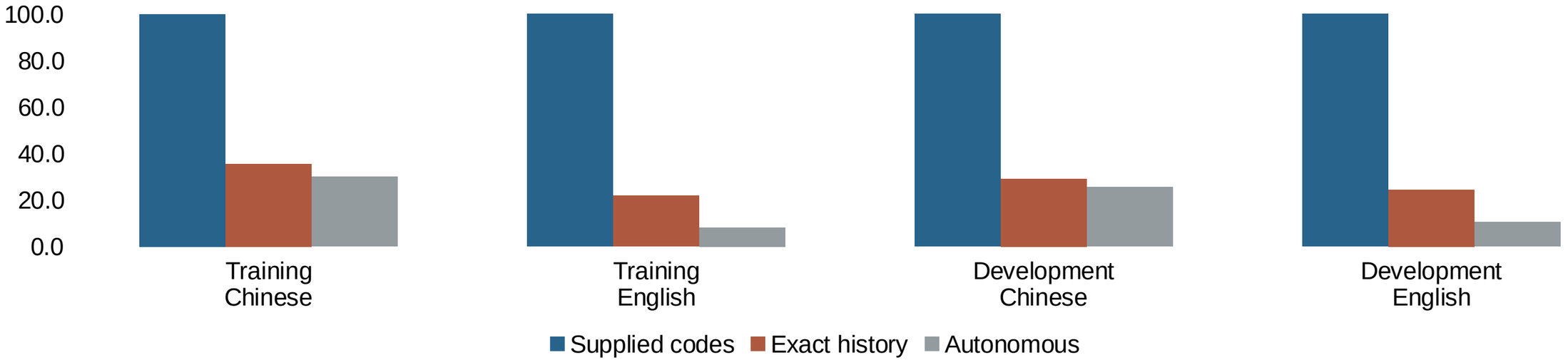
c First two fixed Chinese cases | complete, unmodified streams

	Reference	Full prompt	Blank question
1	玛丽莲·曼森	[illegible]	[illegible]
2	批评他的人	[illegible]	[illegible]

All 384 generations and original gates are verified. Question dependence and stopping pass; answer quality fails.

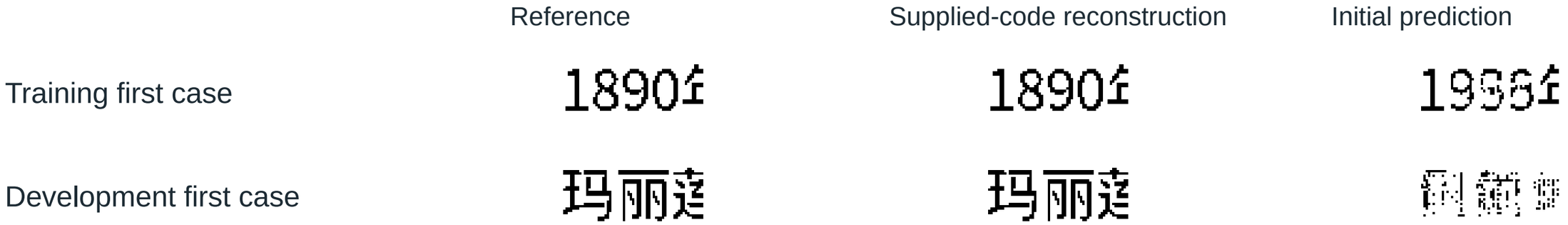
Prediction fails before generated-history drift

a Foreground F1 (%) | eight fixed cases per group



Exact history supplies previous images and answer length; it is an assisted diagnostic.

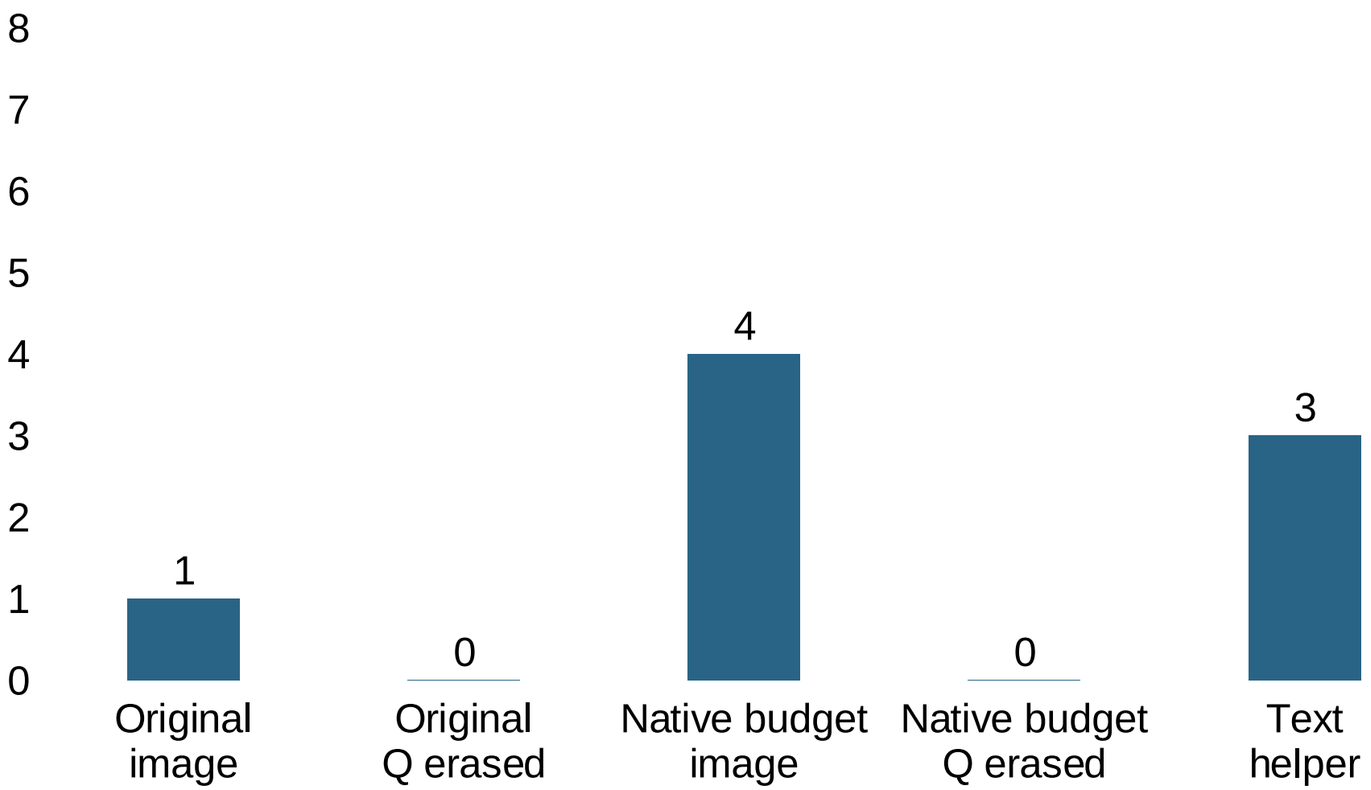
b First 32-by-64 block, before generated history



No first predicted block is exact. All 32 initial predictions match the autonomous path; no diagnostic weights update.

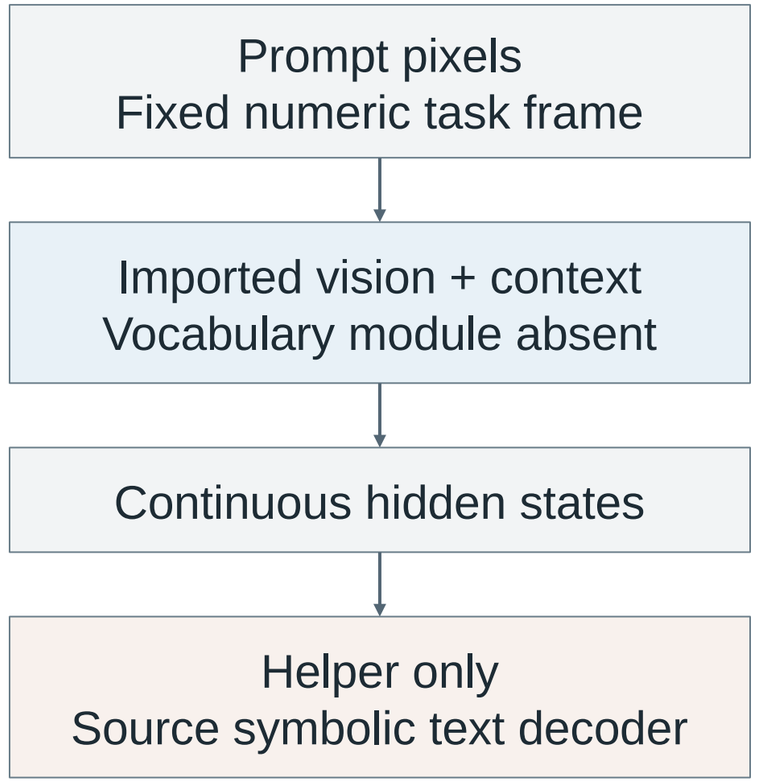
Numeric image controls retain bounded reading

a Exact Chinese answers | the same eight reused questions



Q erased: only question ink is removed, with wrap coordinates retained.

b Diagnostic boundary

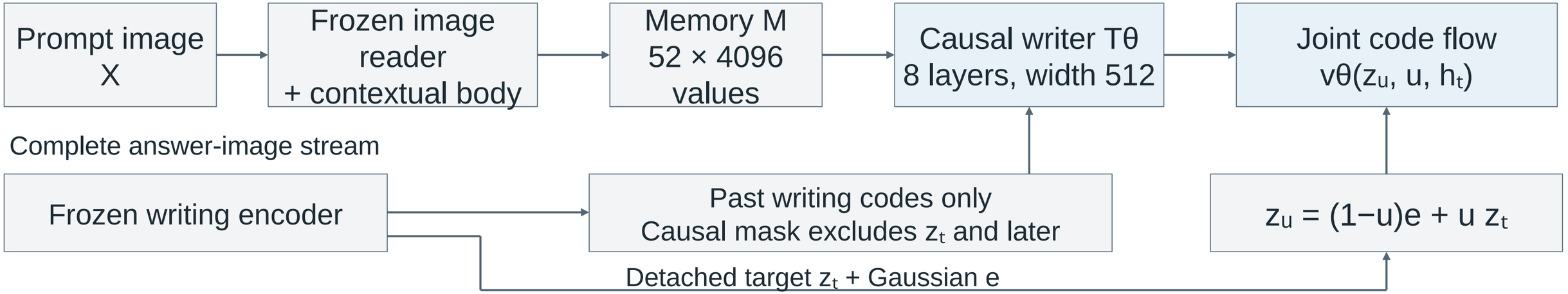


Text decoding is excluded from the image actor.
This check generates no image answers.

All 32 input images and 742 retained decoding choices verify. The complete neural forward pass is not independently replayed.

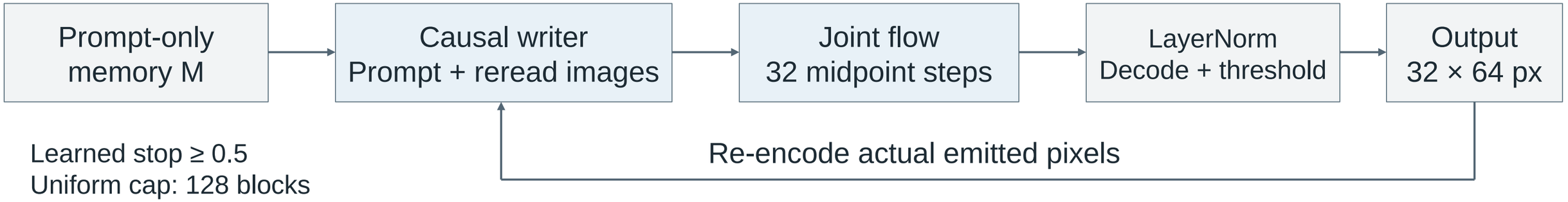
Causal image writing with frozen visual semantics

a Training | complete answer images supervise continuous writing



$$L = \text{mean } ||v\theta - (z_t - e)||^2 + \text{balanced stop loss}$$

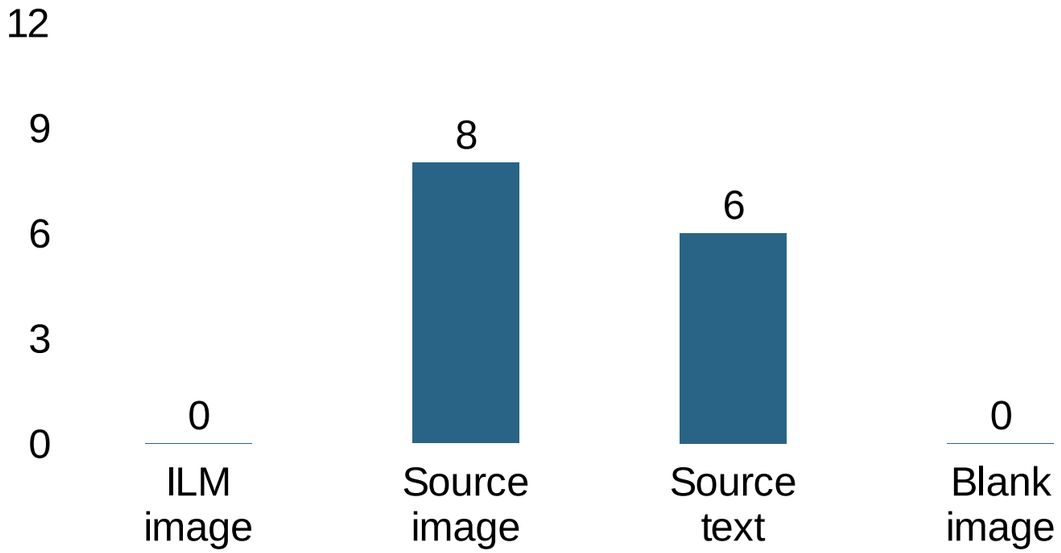
b Inference | every generated block becomes physical visual history



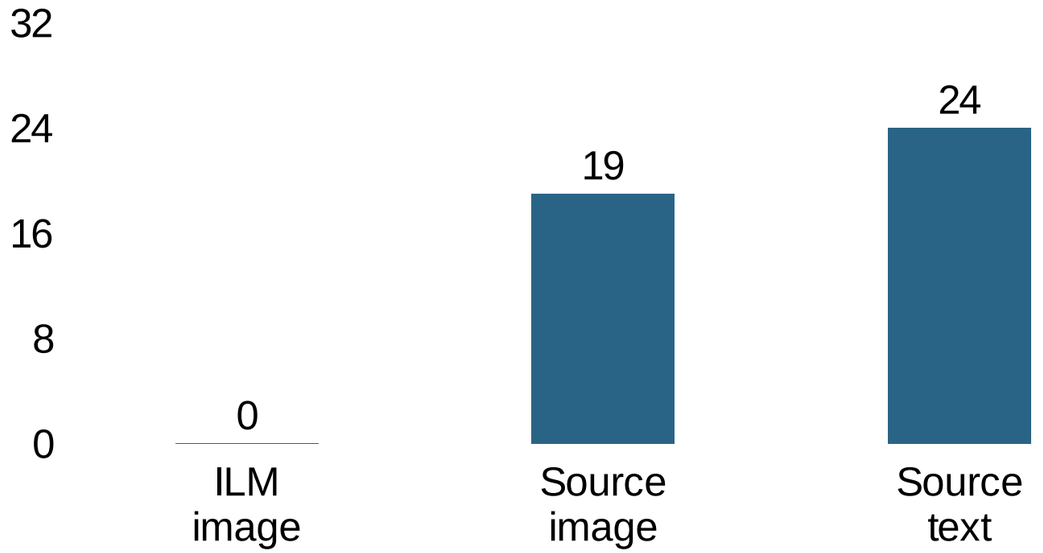
38.65M trainable writer. External reader pretraining retained. Completed Chinese audit: 0/128 exact; 29.386% foreground F1.

Reader content before image writing

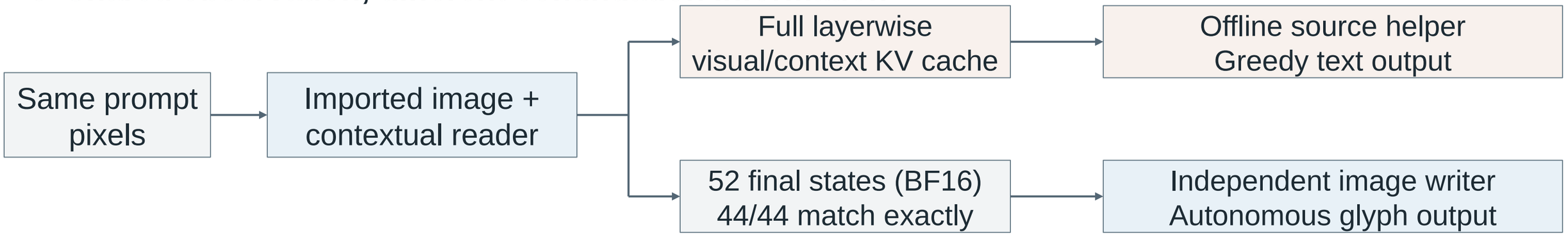
a Exact copied answers / 12 new requests



b Exact answers / 32 fixed training questions



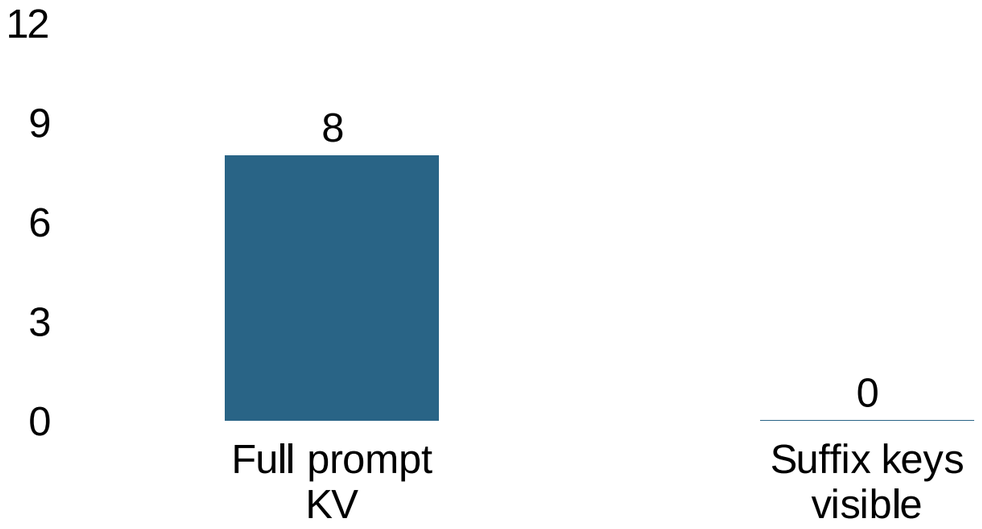
c Shared observation; different continuation information



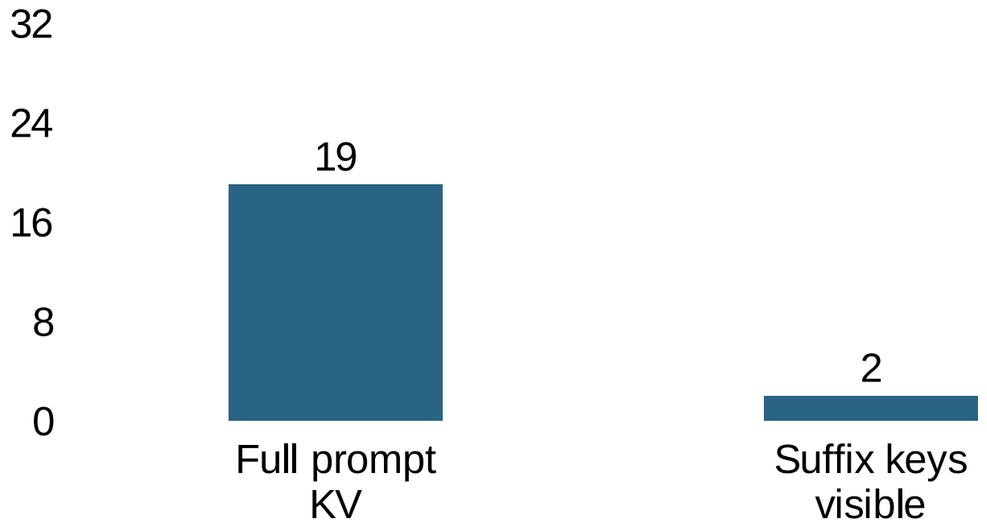
100 fixed offline helper calls. Full-KV success does not prove compact-memory sufficiency or successful ILM image generation.

Context visibility and native image continuation

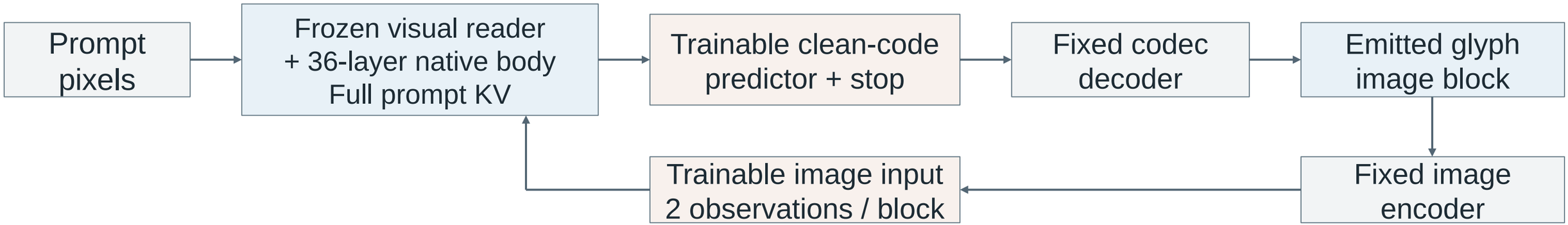
a Exact copy answers / 12 requests



b Exact answers / 32 training questions



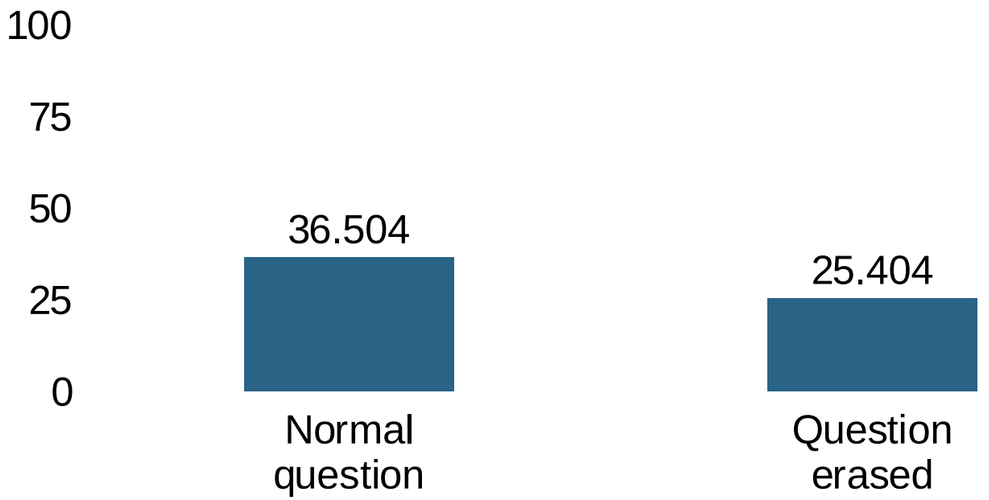
c New image actor: execution only; no trained language result



7.583B total parameters; 46.900M trainable interface parameters. External pretraining retained; 16.638 GiB execution peak.

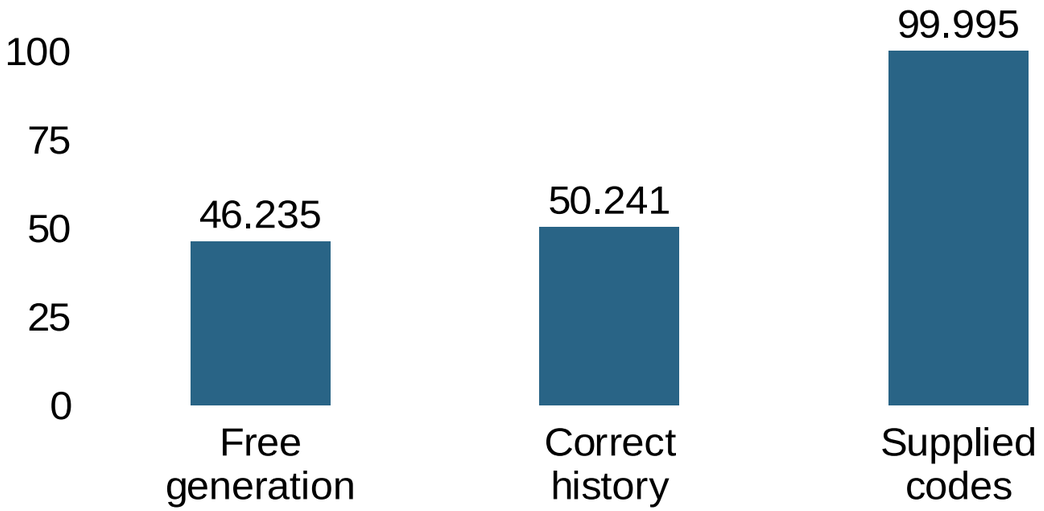
Conditional writing fails after full-context training

a Chinese development / F1 (%)



Exact images: 0/128 in both routes

b Training controls / F1 (%)



Exact images: 0/32, 0/32, 28/32

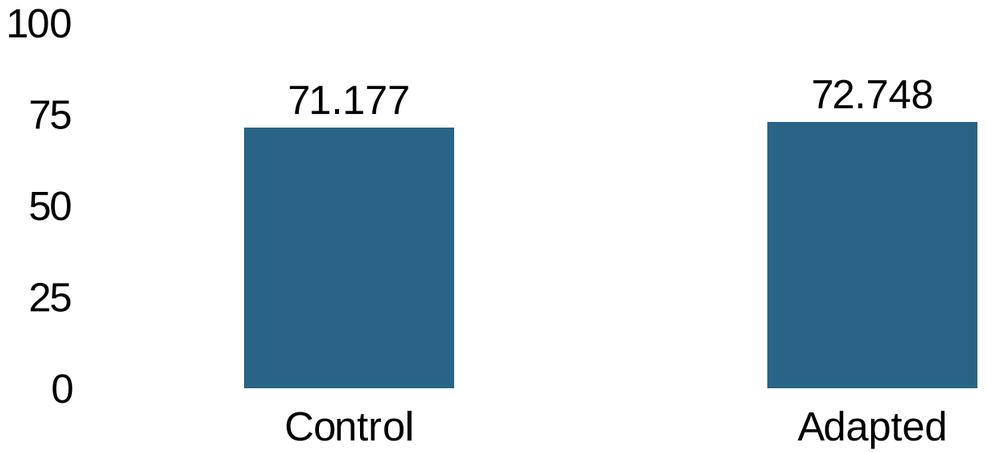
c First fixed question pair / complete raw writing images

	Reference	Free generation	Correct history	Supplied codes
Question A	冯兴			冯兴
Question B	杜英翰			杜英翰

40,000 updates; 7.583B total / 46.900M trainable parameters. Image agreement does not establish semantic accuracy.

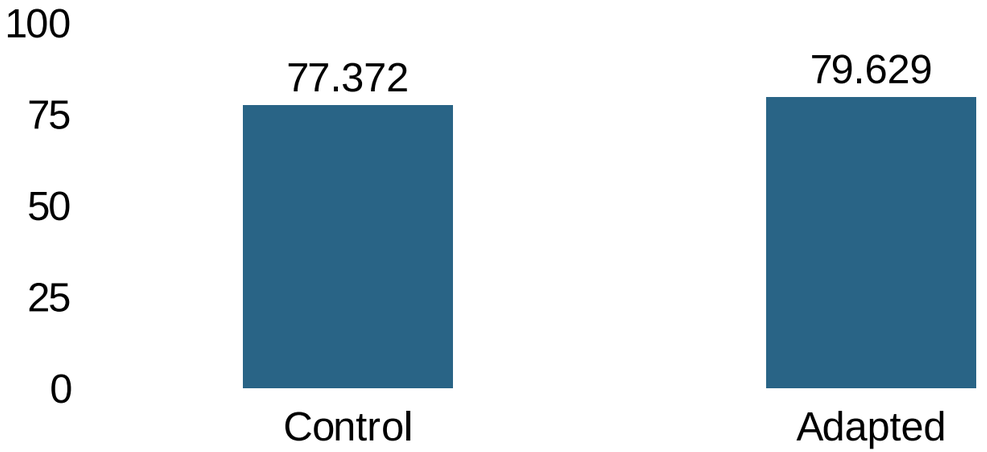
Trainable readout leaves autonomous writing inaccurate

a Autonomous writing / F1 (%)



Exact complete images: 0/32 in both arms

b Correct preceding history / F1 (%)



Exact complete images: 0/32 in both arms

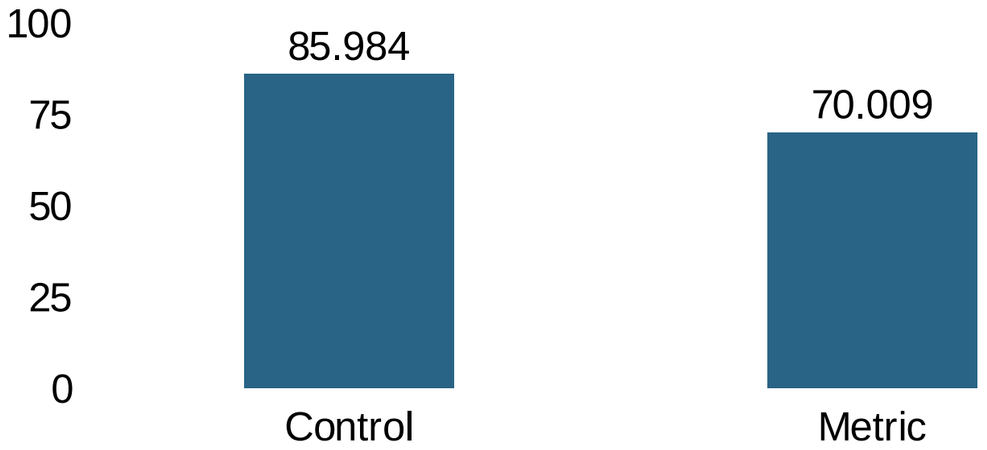
c First four fixed questions / complete raw images

	Reference	Adapted autonomous output
Case 00	冯兴	冯兴
Case 01	杜英翰	杜英翰
Case 02	公元前201年	公元前201年
Case 03	爱琴海沿岸的希腊城邦	爱琴海沿岸的希腊城邦

32 training questions; 2,000 updates/arm. Supplied-code F1: 99.9953% (28/32 exact). Language remains unqualified.

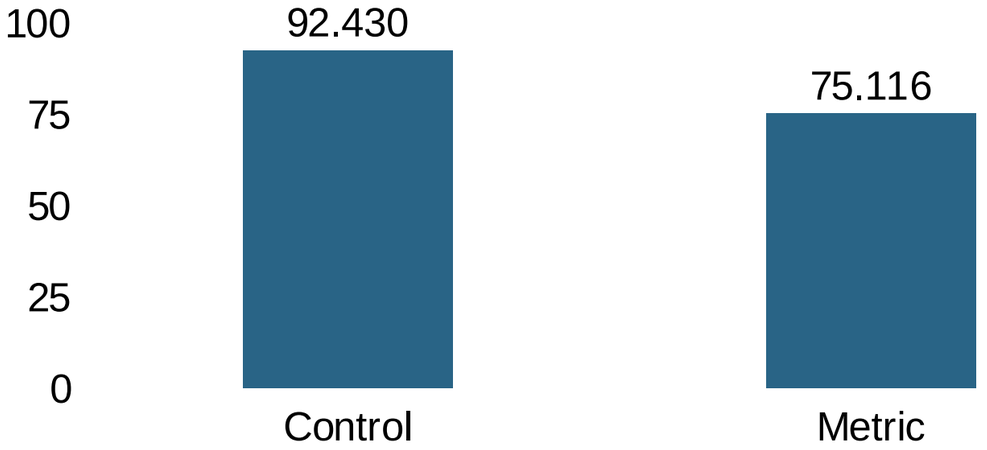
Decoder-sensitive loss reduces answer-image fidelity

a Autonomous writing / F1 (%)



Exact complete images: 0/32 in both arms

b Correct preceding history / F1 (%)



Exact complete images: 0/32 in both arms

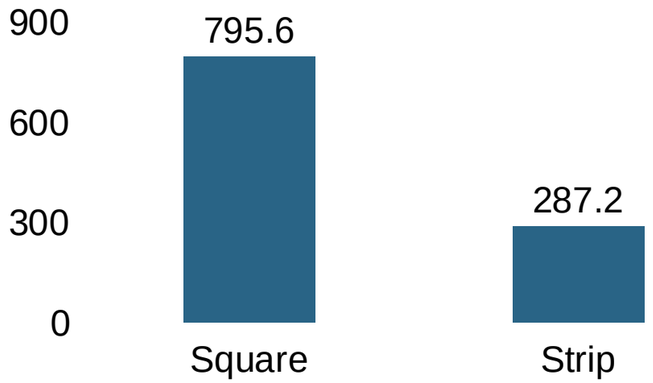
c First four fixed questions / complete raw images

	Reference	Control	Decoder metric
00	冯兴	冯兴	冯兴
01	杜英翰	杜英翰	杜英翰
02	公元前201年	公元前201年	公元前
03	爱琴海沿岸的希腊城邦	爱琴海沿岸的希腊城邦	爱琴海

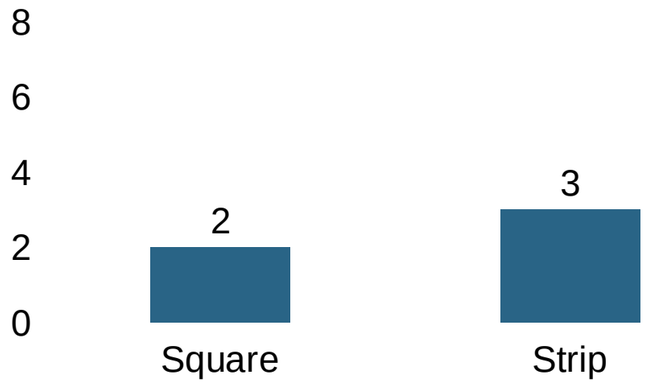
32 training questions; 2,000 updates/arm. All four repair gates fail. Supplied-code F1: 99.9953% (28/32 exact).

Paired image learning remains unreliable

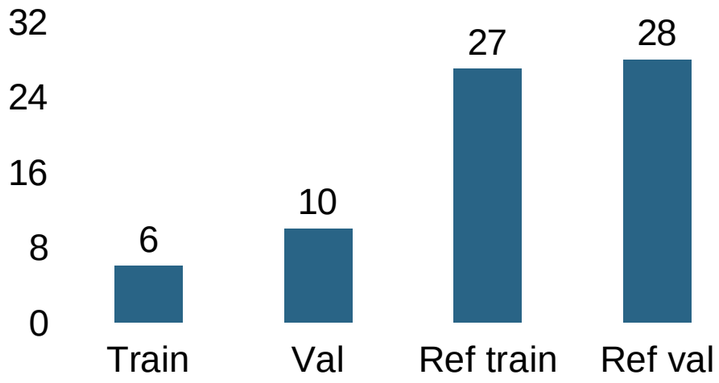
a Training invocation (s)



b Fixed screen: OCR exact / 8



c Full strip: OCR exact / 32



d First four validation cases / full reference and generated canvases

Validation 00

Reference



Generated



Validation 01

Reference

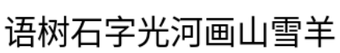


Generated

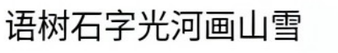


Validation 02

Reference



Generated



Validation 03

Reference



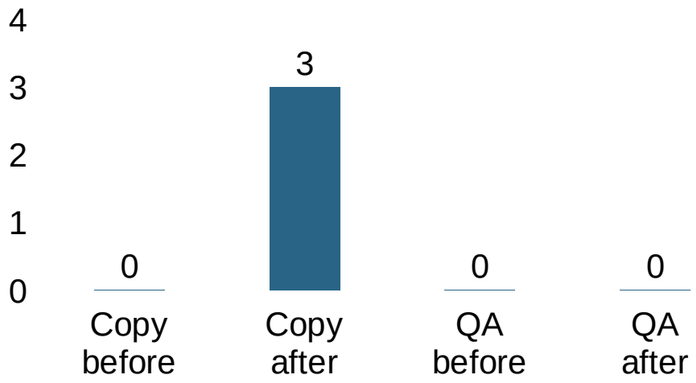
Generated



512 updates/arm on 32 copies; reference OCR: 27/32 train, 28/32 validation. No QA training or language-efficiency claim.

Visual reading after adapting the image frontend

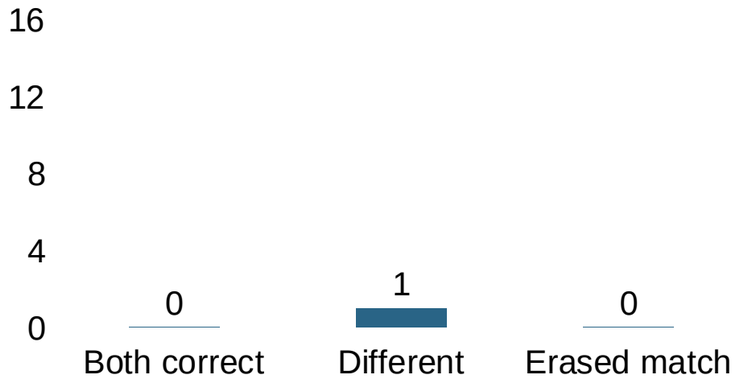
a Held-out screen: exact / 4



b Full validation: exact / 32



c Passage-pair controls / 16



d Source writing: supplied-string controls

花多地地山



mountain bright blue cloud



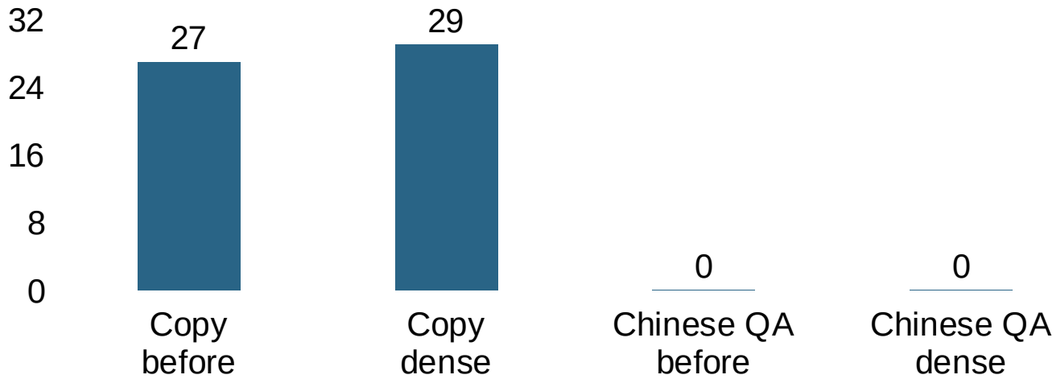
e First four held-out copies: auxiliary text readouts

Requested	After learning
树田红星风	树田红星风
garden silver open blue	garden silver open blue
语树石字光河画山雪羊	语树石字光河画山雪字
green blue	green blue

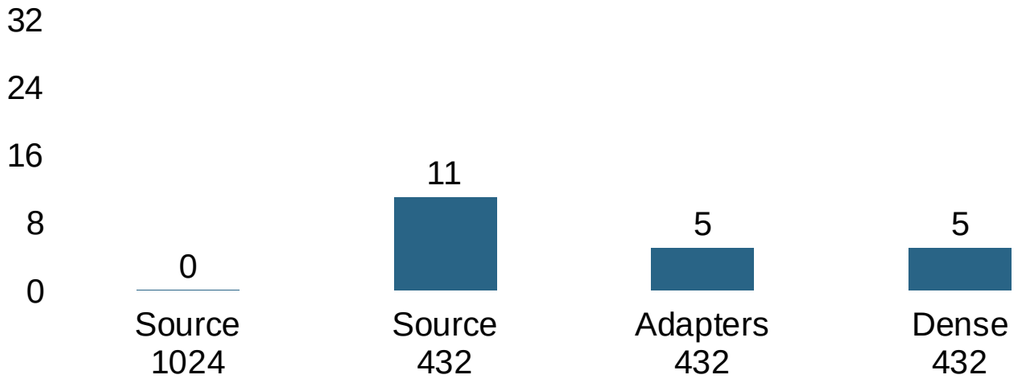
4,096 updates; 384 training examples; only visual adapters learned. Text readouts are not image answers. General ILM remains unachieved.

Native image geometry recovers source reading

a Original geometry: exact answers / 32



b English geometry check: exact / 32



c Same complete page at both input scales

1024 square: 4096 visual positions

Passage:
The programme's first serial, An Unearthly Child, shows that the Doctor has a granddaughter, Susan Foreman. In the 1967 serial, Tomb of the Cybermen, when Victoria Waterfield doubts the Doctor can remember his family because of, "being so ancient", the Doctor says that he can when he really wants to—"The rest of the time they sleep in my mind". The 2005 series reveals that the Ninth Doctor thought he was the last surviving Time Lord, and that his home planet had been destroyed; in "The Empty Child" (2005), Dr. Constantine states that, "Before the war even began, I was a father and a grandfather. Now I am neither." The Doctor remarks in response, "Yeah, I know the feeling." In "Smith and Jones" (2007), when asked if he had a brother, he replied, " No, not any more." In both "Fear Her" (2006) and "The Doctor's Daughter" (2008), he states that he had, in the past, been a father.
Question: What is the name of the first Doctor Who serial?
Answer:

432 square: 729 visual positions

Passage:
The programme's first serial, An Unearthly Child, shows that the Doctor has a granddaughter, Susan Foreman. In the 1967 serial, Tomb of the Cybermen, when Victoria Waterfield doubts the Doctor can remember his family because of, "being so ancient", the Doctor says that he can when he really wants to—"The rest of the time they sleep in my mind". The 2005 series reveals that the Ninth Doctor thought he was the last surviving Time Lord, and that his home planet had been destroyed; in "The Empty Child" (2005), Dr. Constantine states that, "Before the war even began, I was a father and a grandfather. Now I am neither." The Doctor remarks in response, "Yeah, I know the feeling." In "Smith and Jones" (2007), when asked if he had a brother, he replied, " No, not any more." In both "Fear Her" (2006) and "The Doctor's Daughter" (2008), he states that he had, in the past, been a father.
Question: What is the name of the first Doctor Who serial?
Answer:

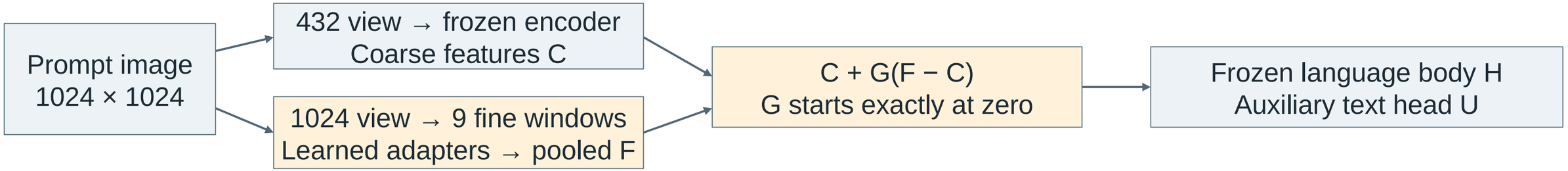
d First four source answers at 432: text readouts

Primary reference	Source answer
An Unearthly Child	The name of the first Doctor Who serial is An Unearthly Child.
Susan Foreman	SUSAN FOREMAN
7.5	7.5%
32	32%

Auxiliary Unicode readouts. Exact scores accept all annotated alternatives and require EOS. No trained image-answer result.

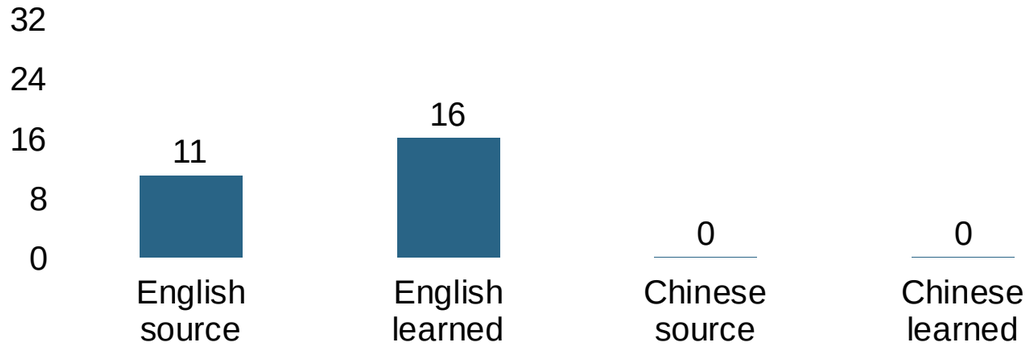
Visual-state learning and canvas dependence

a Learned correction to a frozen native reader



Training only: original-text pre-answer states at seven layers; answer CE + normalized state MSE

b Held-out QA: exact stopped answers / 32



Copying: English 7/16 → 16/16; Chinese 0/16 → 0/16

c Separate dense source: Chinese copies / 16



English copies: 16/16 → 7/16 → 0/16; QA stays 0/32

d Seen copying request; white display padding omitted

请只抄写引号内的内容：
“花多地地山”

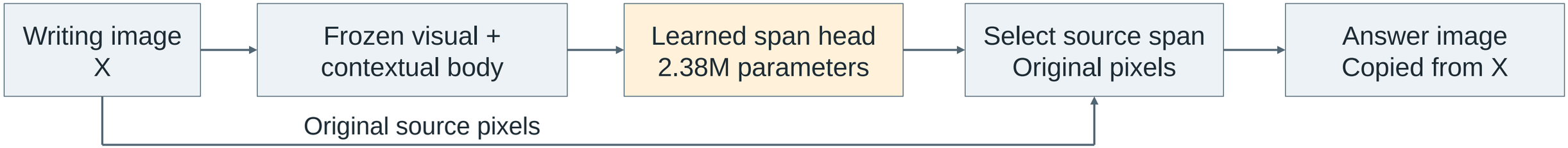
Raw auxiliary text readout (not an answer image)

Requested: 花多地地山
Learner: 花多梅树山
All 8 tested seen Chinese strings change characters

State correction: 4/6 requirements fail.
Attention-only: QA 0/32 both languages; 6/6 fail.

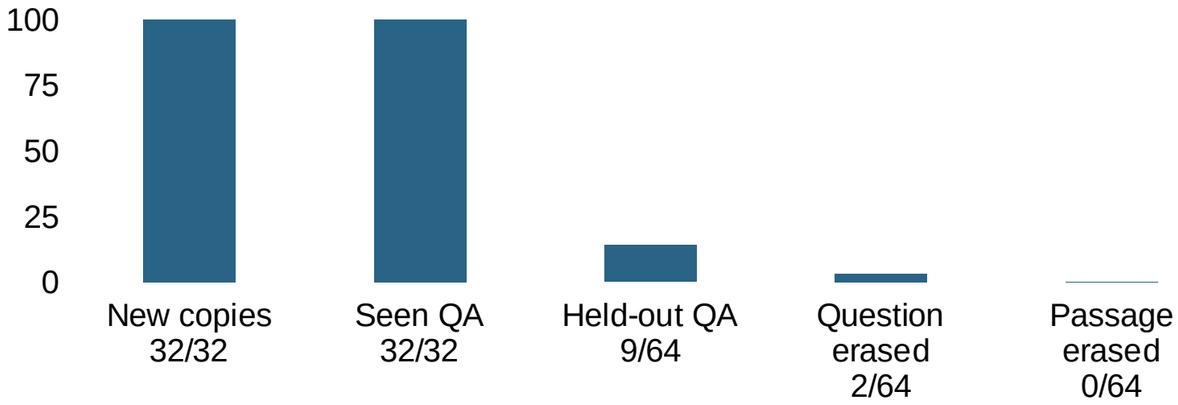
Exact copying with weak question answering

a Image-only contextual span selection



Normalized scores over visible spans; duplicate images share probability mass

b Exact image answers (%); counts below



Both questions correct: 1/32 paragraph pairs

3/4 requirements fail; source-copy component only

c First fixed held-out answer pair

1. What are the fish's common names?

Reference 1

国公、白虎

Output 1: wrong

为辐鳍鱼纲鲀形目鲀亚目，旧属鲀科

2. Which ocean regions does it inhabit?

Reference 2

印度洋及西太平洋区

Output 2: correct

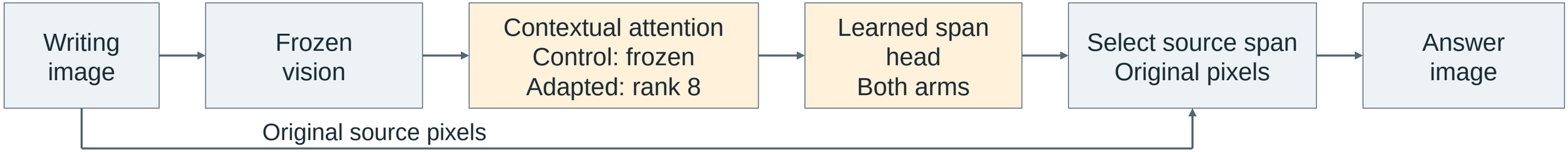
印度洋及西太平洋区

7.531B full runtime with external pretraining; 320 training images, 3,200 updates

Source copies cannot write answers absent from the input; free generation remains unachieved

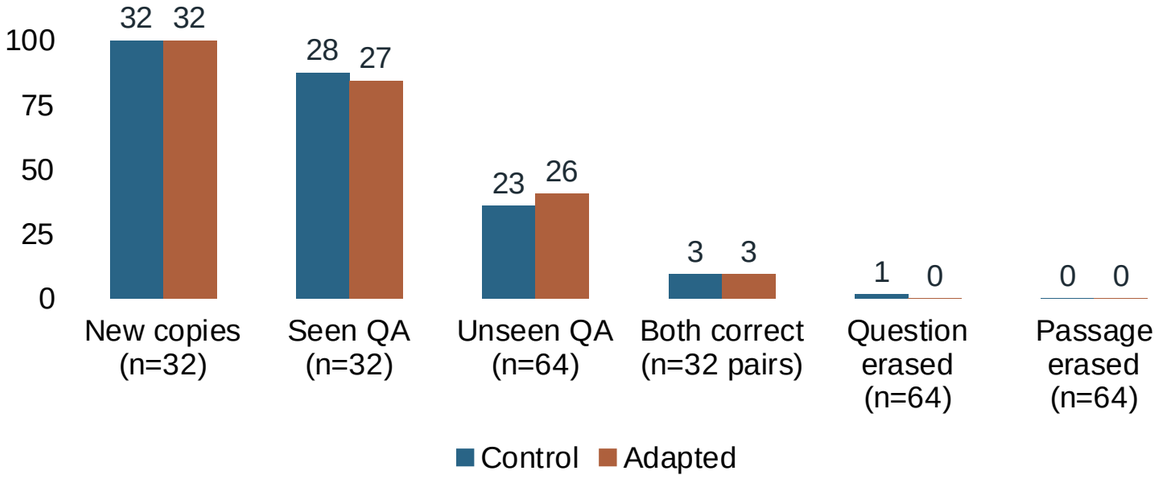
Contextual learning improves few answers and misses reader requirements

a Same source-pixel motor; matched learning schedule



3,257 updates per arm; 2.378M / 8.572M learned parameters; 7.531B pretrained full runtime

b Exact image answers (%); labels give correct counts



Gain: +3 answers, +0 complete pairs

Both fail reader floors; adaptation misses improvement rule

c First fixed unseen question pair

1. What are the fish's common names?

Reference	国公、白虎
Control	国公、白虎，
Adapted	国公、白虎，

2. Which ocean regions does it inhabit?

Reference	印度洋及西太平洋区
Control	印度洋及西太平洋区
Adapted	印度洋及西太平洋区

Both add a comma in Q1; both reproduce Q2 exactly

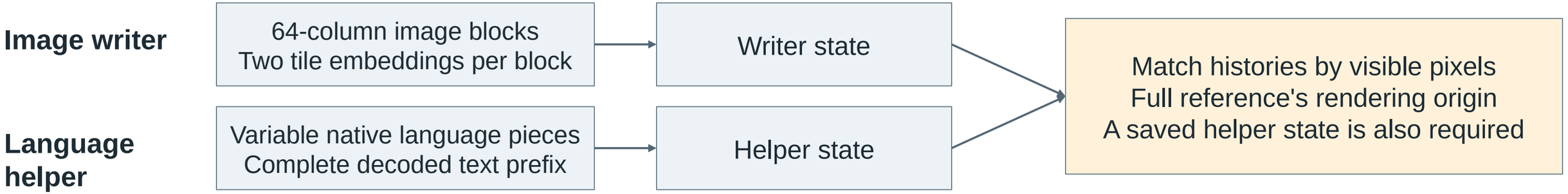
d Separate task-memory substitution: earlier head

53/64 development selections change; original score 9/64. Hybrid features, not native new prompts.

Source crops cannot compose absent answers. Earlier-head substitution is not a third comparison arm.

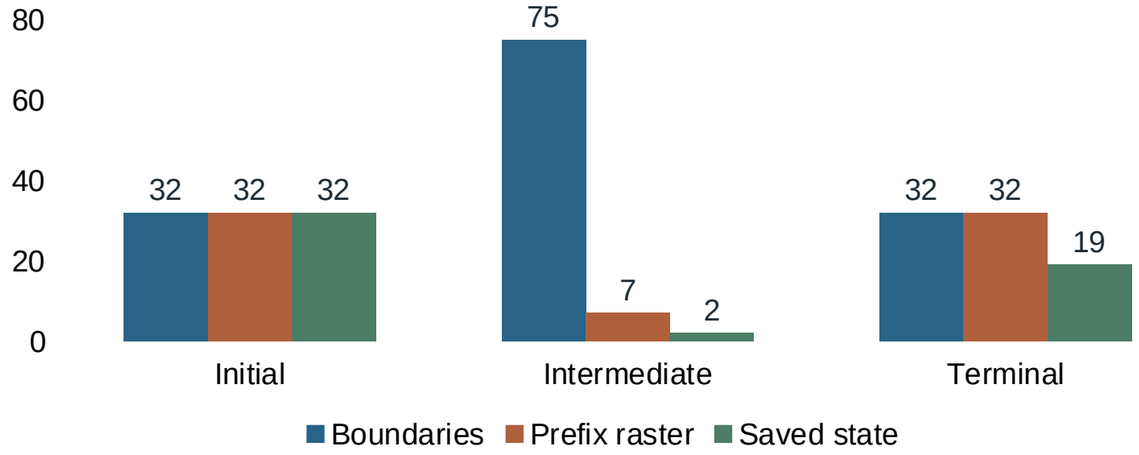
Image steps and language pieces usually describe different histories

a Audit identical preceding content before comparing states



Equal ordinal step does not establish equal history

b All retained boundary counts



All 32 initial states agree after common BF16 rounding

c Both eligible intermediate matches

TRAIN_1779_QUERY_1

美国维吉尼亚州的

Image positions: 6 Helper positions: 6

Normalized state distance: 1.497 (uncalibrated)

TRAIN_2146_QUERY_1

1.7-2.

Image positions: 2 Helper positions: 6

Normalized state distance: 1.407 (uncalibrated)

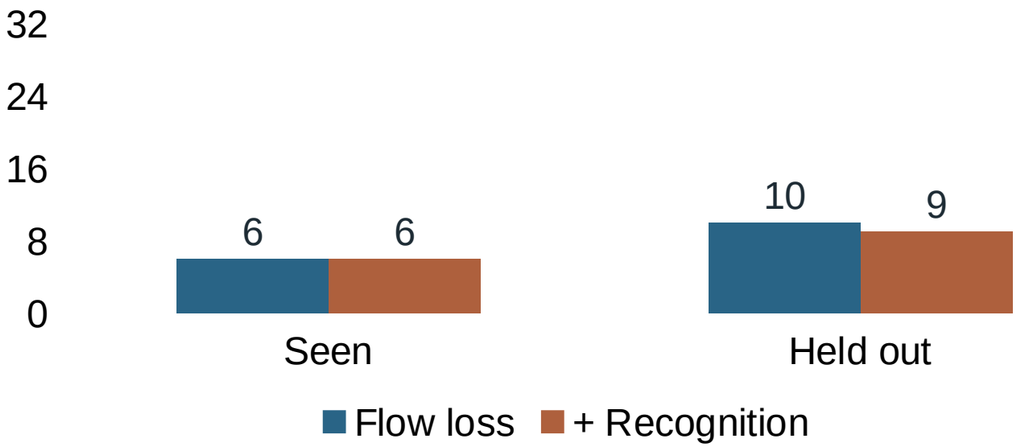
68/75 intermediate boundaries have no complete-prefix raster; 5 more lack a saved matching state

All 216 saved source-prefix positions decode to complete UTF-8; 13 terminal states are missing

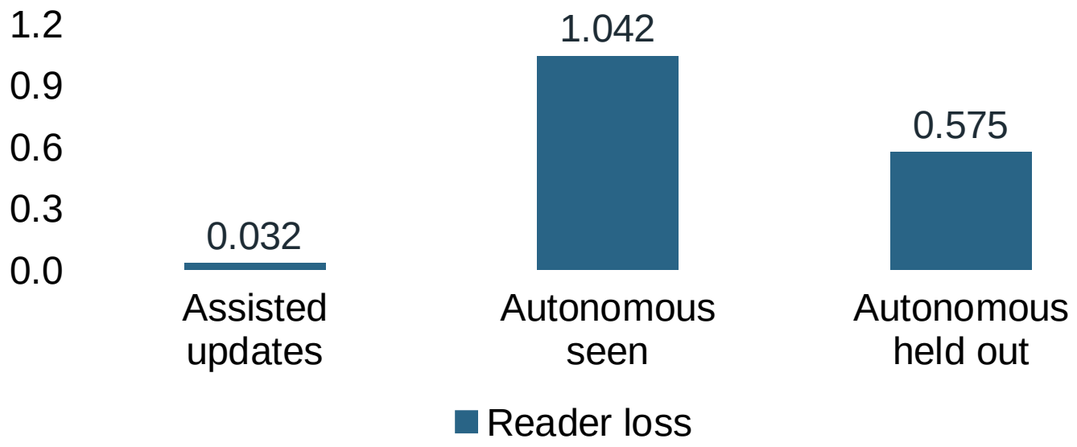
Existing traces only: zero new neural calls, zero updates. No alignment treatment or writer repair tested.

Recognition loss does not ensure faithful visible writing

a Independent OCR-exact copies (32 per split)



b Training-reader target NLL (nats)



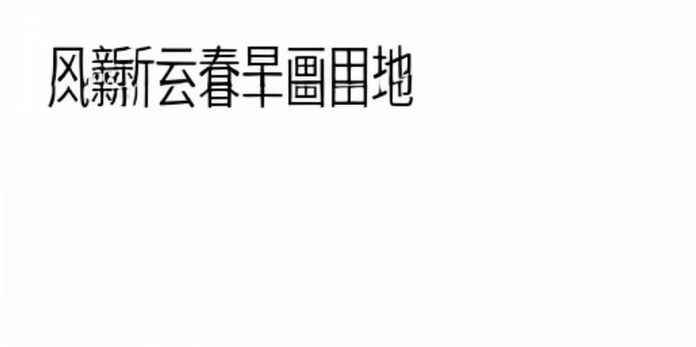
255 assisted updates and final-state free outputs are different regimes; this is not a causal decomposition.

c Reader ignores a visible duplicated line

Reference

风新云春早画田地

Actual generated page



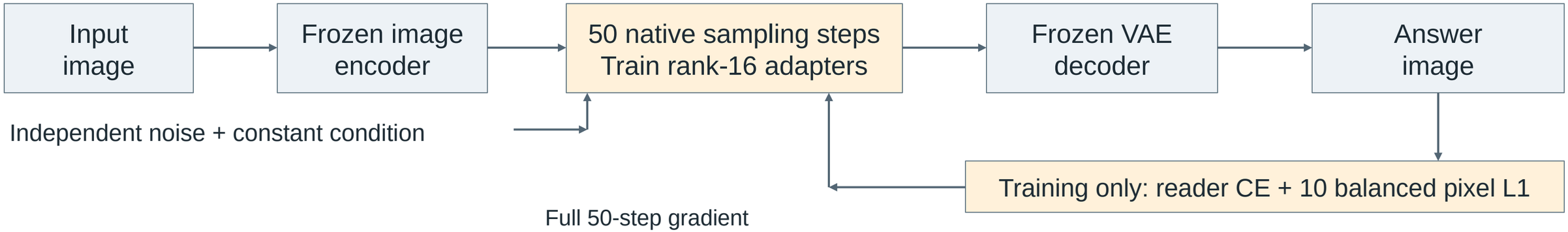
Reader returns the expected string.
Gold NLL: 0.0424; clean reference: 0.000192.
Recognition does not count every visible repetition.

6/32 and 9/32 with the independent evaluator; 14/32 and 19/32 with the training reader.
The reader is not an independent judge. No reliable writer or semantic QA is established.

Same first 32 image pairs; 512 updates; source backbone and condition fixed. Complete 64-case outputs retained.

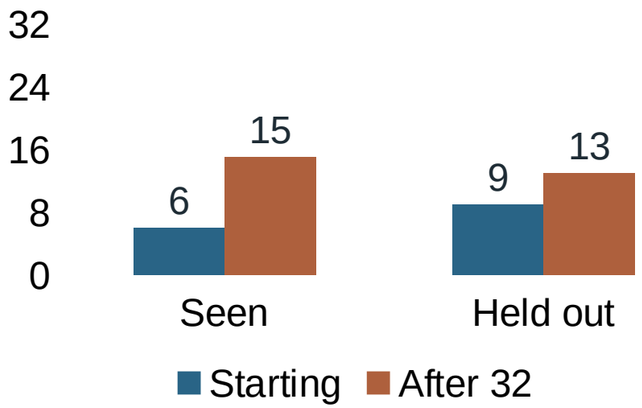
Autonomous image-to-image learning remains below reliable writing

a Full sampling trajectory; targets enter only the final loss



32 updates: 1,135.452 s; 12.057 GiB peak allocated. Runtime: 3.900B transformer + 84.046M VAE.

b OCR-exact copies (n=32)



Progress rule met; 30/32 writing floor not met.
No matched extra-training control.

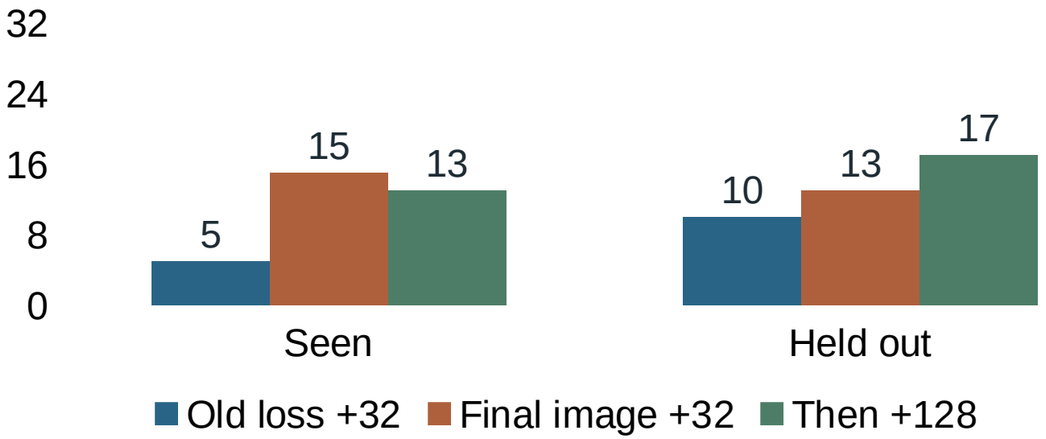
c First two held-out inputs: actual pages

Target	Starting	After pilot
树田红星风	树田丝红星风”	树田红工星风”
garden silver open blue	garden silvver ope blue	garden silver open blue

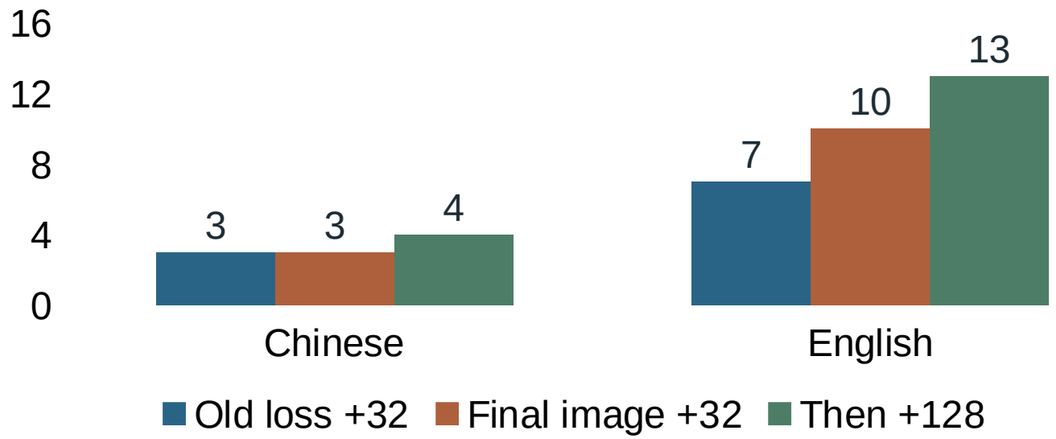
Unwanted punctuation remains in Chinese; the English phrase restores its missing letter.

Balanced exposure and autonomous image writing

a Complete OCR-exact copies (32 per split)



b Held-out copying by language (16 each)



128 additional updates
Four complete passes over 32 examples
64 Chinese and 64 English draws

Training: 4,500.1 s
Peak: 12.032 GiB
Full 50-step trajectory gradient

Continuation rule: not met
Copying does not establish semantic QA.
Extra updates and exposure change together.

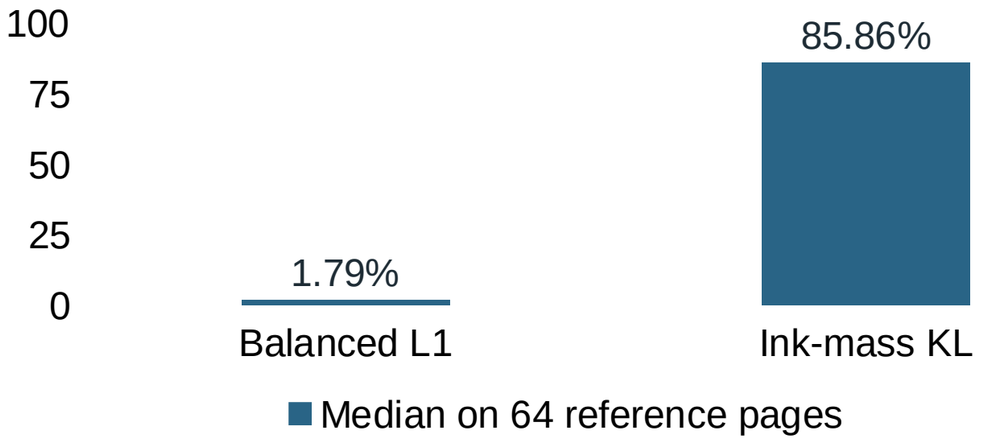
c First two held-out inputs: complete actual pages

Target	Starting pilot	After +128
树田红星风	树田红工星风”	树田红工星风”
garden silver open blue	garden silver open blue	garden silver open blue

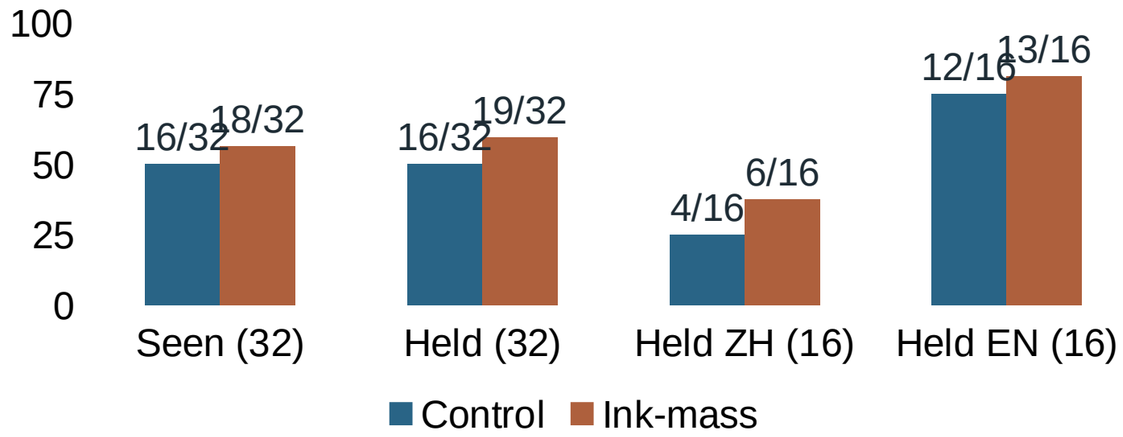
Canonical writing style; reused development cases; inherited 3.900B transformer + 84.046M VAE.

Fixed target ink mass improves some copies

a Duplicate / deletion physical cost (%)



b Complete OCR-exact copies (%)



32 matched updates; same input order
Same weights, Adam state and noise
Only physical loss changes
Reader CE is unchanged

KL scale: 0.403568; temperature: 0.05
Fixed target ink mass; unclipped RGB
Training peaks: 12.032 / 12.033 GiB

Exploratory rule passes; writing floor fails.
CPU corruptions are not model outputs.
No semantic QA in these two arms.

c Fixed first two held-out cases: complete actual pages

Target	Control	Ink-mass
树田红星风	树田红工星风”	树田红星风
garden silver open blue	garden silver open blue	garden silver open blue

Single seed; reused development; inherited 3.900B transformer + 84.046M VAE; all 128 final outputs retained.

Full pages survive encoding; answering remains untrained

a Complete first input and native reconstruction

Original 1024 x 1024 input

资料：大唐威龙，是香港漫画《天子传奇》系列的第四部作品，讲述的故事是唐太宗李世民如何成为天子的故事。大唐威龙是黄玉郎作品当中，评价相当高的一部、甚至于有人认为是最好的一部。故事以奇情开场，描述李世民其实是双胞胎，大孖命格富贵，但是短命。细孖虽然只晚一刻钟出生，但有帝王之相。于是极乐正宗的宗主摩诃叶，便将命带帝王相的细孖掳走，秘密训练李世民武功六神诀，准备扶持李世民登上皇位，自己高居太上皇。李世民受命将孪生大哥杀死（应了短命的命数）后，混入李家开始江湖历练。此时唯一目击杀人顶替过程的正一道道士李淳风，则被摩诃叶下了格杀令四处遁逃。故事遂采双线并行。到了杨玄感觉醒霸王转世宿命后，又转为三位主角的三分线。
问题：大唐威龙，是香港漫画《天子传奇》系列的第几部作品?
回答：

Frozen native VAE reconstruction

资料：大唐威龙，是香港漫画《天子传奇》系列的第四部作品，讲述的故事是唐太宗李世民如何成为天子的故事。大唐威龙是黄玉郎作品当中，评价相当高的一部、甚至于有人认为是最好的一部。故事以奇情开场，描述李世民其实是双胞胎，大孖命格富贵，但是短命。细孖虽然只晚一刻钟出生，但有帝王之相。于是极乐正宗的宗主摩诃叶，便将命带帝王相的细孖掳走，秘密训练李世民武功六神诀，准备扶持李世民登上皇位，自己高居太上皇。李世民受命将孪生大哥杀死（应了短命的命数）后，混入李家开始江湖历练。此时唯一目击杀人顶替过程的正一道道士李淳风，则被摩诃叶下了格杀令四处遁逃。故事遂采双线并行。到了杨玄感觉醒霸王转世宿命后，又转为三位主角的三分线。
问题：大唐威龙，是香港漫画《天子传奇》系列的第几部作品?
回答：

Four full pages; features: 128 x 64 x 64
Foreground F1: 97.834-98.180% (physical reconstruction)

First input question region (display inset only)

问题：大唐威龙，是香港漫画《天子传奇》系列的第几部作品?
回答：

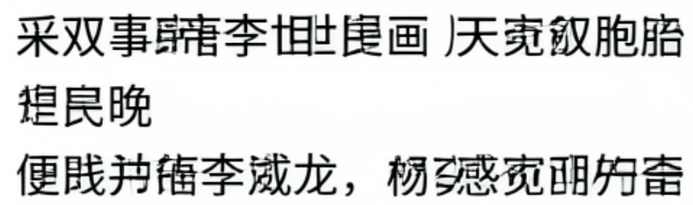
Two real QA updates discarded: 12.203 GiB. Fresh inference reloads the fixed earlier checkpoint.
0/4 normal answers; eight raw normal/control pages retained. Small preflight, not language learning.

b First two natural questions Unoptimized baseline

Q1 reference (whitespace cropped)

第四部

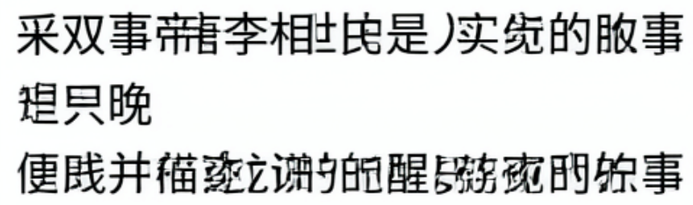
Actual output: complete 512 x 256 page



Q2 reference (whitespace cropped)

唐太宗李世民如何成为天子

Actual output: complete 512 x 256 page



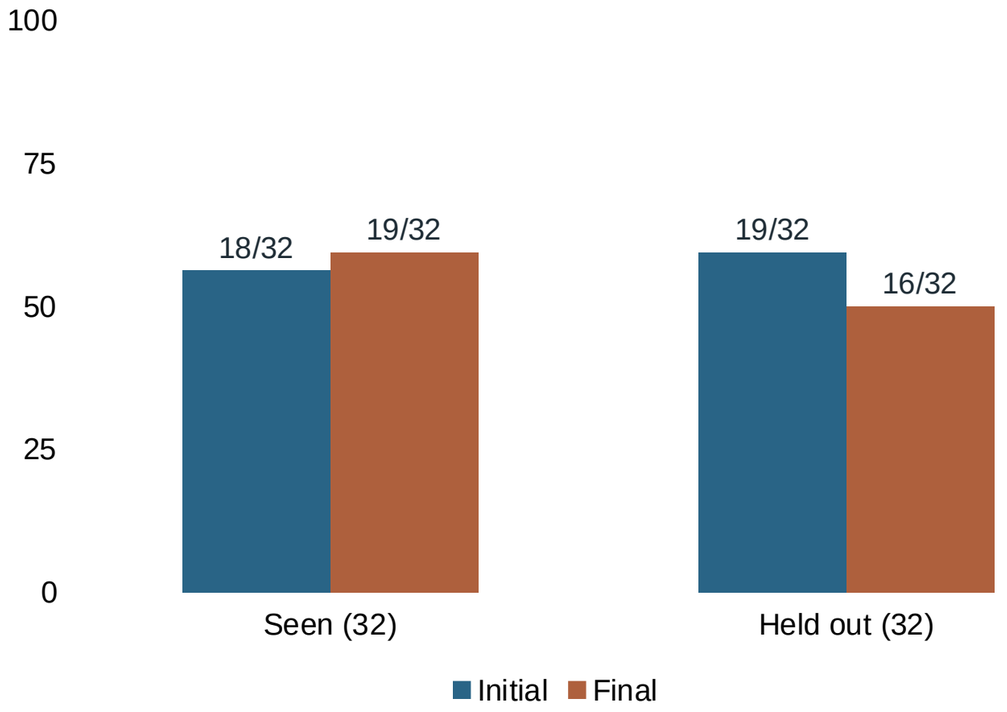
Natural-QA adaptation does not establish answering

a Source-accepted OCR-exact answers

Complete cohort	Initial	Final
Seen questions	0/32	0/32
Held-out questions	0/32	0/32
Held-out pairs, both correct	0/16	0/16
Held out, question erased	0/32	0/32
Held out, passage erased	0/32	0/32

All six prospective conditions fail.
Held-out copying loses three exact strings; at most two allowed.

b Complete OCR-exact copies (%)



128 updates on 32 questions from 16 passages
3.55 h training; 12.204 GiB allocated peak
24.359M trained adapters in a 3.900B image transformer

384 original answers reviewed; reference OCR: 23/32 seen, 26/32 held. Reused public data; one schedule; inherited pretraining.

Fixed held-out pair: complete unrepaired answer pages

a Complete question inputs

Question 1

资料：四线天竺鲷（学名：），为辐鳍鱼纲鲈形目鲈亚目天竺鲷科天竺鲷属的一种。本鱼分布于印度西太平洋区，包括马达加斯加、留尼旺、塞席尔群岛、斯里兰卡、台湾、泰国、越南、印度、日本、马来西亚、印度尼西亚、菲律宾以及中国南海及东海、澳洲、斐济等海域。该物种的模式产地在印度Pondicherry。水深0至80公尺。本鱼体延长而侧扁，眼大，口大略下位。鱼体黄褐色，腹部银白色，体侧具4条深色细纵带，其中一条从吻部穿过眼睛延伸至尾鳍底。背鳍硬棘8枚；背鳍软条9枚；臀鳍硬棘2枚；臀鳍软条8枚，体长可达13公分。本鱼栖息于珊瑚礁或岩礁区，白天躲藏于岩架下或岩洞中，夜间出来觅食。繁殖期时，雄鱼具有口孵习性，卵约7日化成仔鱼，由雄鱼吐出，具短暂的仔鱼飘浮期。不具经济价值。
问题：四线天竺鲷一般分布在什么地方？
回答：

Question 2

资料：四线天竺鲷（学名：），为辐鳍鱼纲鲈形目鲈亚目天竺鲷科天竺鲷属的一种。本鱼分布于印度西太平洋区，包括马达加斯加、留尼旺、塞席尔群岛、斯里兰卡、台湾、泰国、越南、印度、日本、马来西亚、印度尼西亚、菲律宾以及中国南海及东海、澳洲、斐济等海域。该物种的模式产地在印度Pondicherry。水深0至80公尺。本鱼体延长而侧扁，眼大，口大略下位。鱼体黄褐色，腹部银白色，体侧具4条深色细纵带，其中一条从吻部穿过眼睛延伸至尾鳍底。背鳍硬棘8枚；背鳍软条9枚；臀鳍硬棘2枚；臀鳍软条8枚，体长可达13公分。本鱼栖息于珊瑚礁或岩礁区，白天躲藏于岩架下或岩洞中，夜间出来觅食。繁殖期时，雄鱼具有口孵习性，卵约7日化成仔鱼，由雄鱼吐出，具短暂的仔鱼飘浮期。不具经济价值。
问题：四线天竺鲷的模式产地是哪里？
回答：

Q1: source question inset (display only)

问题：四线天竺鲷一般分布在什么地方？
回答：

Q2: source question inset (display only)

问题：四线天竺鲷的模式产地是哪里？
回答：

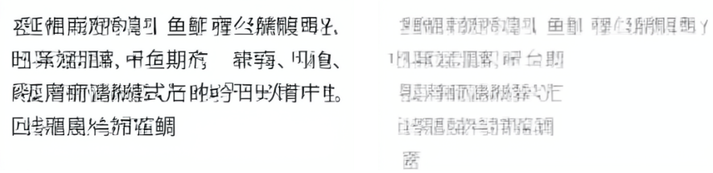
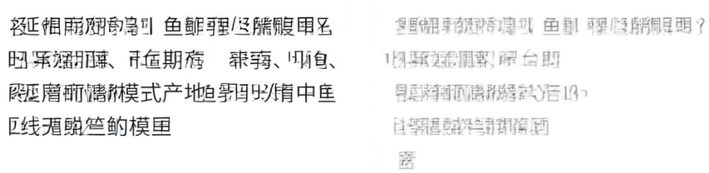
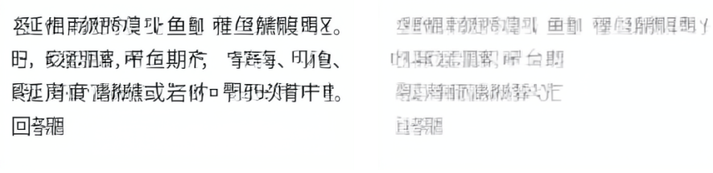
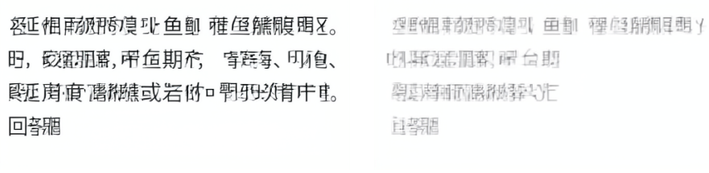
b Original answer pages

Q1 reference (whitespace cropped)

印度西太平洋区

Q2 reference (whitespace cropped)

印度Pondicherry

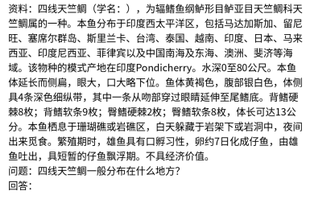
Initial	Final	Initial	Final
Normal			
			
Question erased			
			
Passage erased			
			

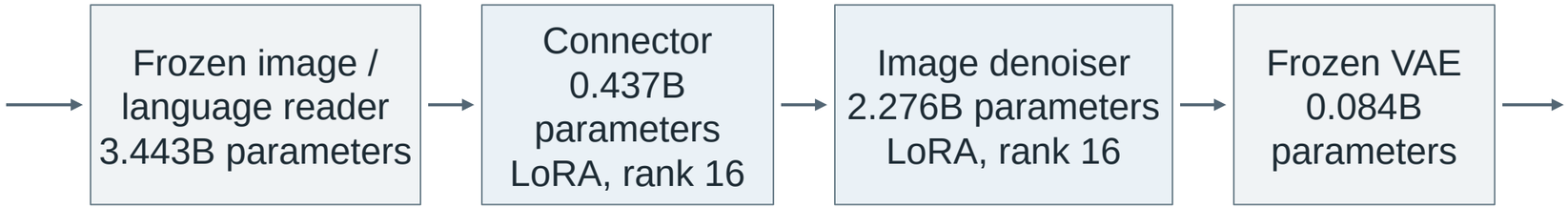
First registered held-out pair, not selected by appearance. All 12 model pages are complete; only input/question and reference display insets are cropped.

Native image-answer flow: trained execution, failed answering

a Actual input and sampling path

Input page x


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问题：四线天竺鲷一般分布在什么地方？
回答：



Actual answer image



Full unrepaired page

1024 × 1024 RGB

Reference latent r(x): same frozen VAE encoder

b Training-only image supervision

Answer image y → frozen VAE encoder E(y)

$z\sigma = (1 - \sigma)\, E(y) + \sigma\, \varepsilon \qquad v^* = \varepsilon - E(y)$

$\text{Loss} = \text{MSE}(v\theta(z\sigma, t, x), v^*)$

One logical denoiser call per update; positive task branch.
Loss updates connector / writer adapters; reader and VAE stay frozen.

c Matched source / trained results

	Initial	Final
Seen questions	0/32	**0/32**
Held-out questions	0/32	**0/32**
Both held-out correct	0/16	**0/16**
Seen copying strings	0/32	**2/32**
Held-out copying strings	0/32	**3/32**

3.18 h training; 15.697 GiB peak
37.381M trained adapters; 6.240B retained runtime
QA reference OCR: 23/32 seen; 26/32 held out
Copy reference OCR: 27/32 seen; 29/32 held out

All 384 fixed initial/final outputs authenticated and inspected. Inherited language/image pretraining. Zero QA counts use source-accepted OCR; useful language learning remains unproved.